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United States Patent Application Publication

20250278020

Kind Code

A1

Publication Date

September 04, 2025

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CHEMICALLY AMPLIFIED NEGATIVE RESIST COMPOSITION AND RESIST PATTERN FORMING PROCESS

Abstract

A chemically amplified negative resist composition comprising (A) a photoacid generator in the form of an onium salt containing a sulfonic acid anion having a substituted fused ring structure and an aromatic sulfonic acid structure and (B) a base polymer is provided. The resist composition exhibits a high resolution during pattern formation and forms a pattern with satisfactory LER and fidelity.

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Appl. No.: 19/064233
Filed: February 26, 2025

Foreign Application Priority Data

JP 2024-031901 Mar. 04, 2024

Publication Classification

Int. Cl.: G03F7/038 (20060101); C09D135/02 (20060101); G03F7/00 (20060101); G03F7/004 (20060101)

U.S. Cl.:

CPC G03F7/0382 (20130101); C09D135/02 (20130101); G03F7/0005 (20130101); G03F7/0045 (20130101); G03F7/0046 (20130101); G03F7/0388 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This non-provisional application claims priority under 35 U.S.C. § 119 (a) on Patent Application No. 2024-031901 filed in Japan on Mar. 4, 2024, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] This invention relates to a chemically amplified negative resist composition and a resist pattern forming process using the same.

BACKGROUND ART

[0003] To meet the demand for higher integration density and operating speed of LSIs, the effort to reduce the pattern rule is in rapid progress. Acid-catalyzed chemically amplified resist compositions are most often used in forming resist patterns with a feature size of 0.2 μm or less. High-energy radiation such as UV, deep-UV or EB is used as the light source for exposure of these resist compositions. In particular, while EB lithography is utilized as the ultra-fine microfabrication technique, it is also indispensable in processing photomask blanks to form photomasks for use in semiconductor device fabrication.

[0004] Polymers comprising a major proportion of aromatic structure having an acidic side chain, for example, polyhydroxystyrene are useful in resist materials for the KrF excimer laser lithography. These polymers are not used in resist materials for the ArF excimer laser lithography because they exhibit strong absorption at a wavelength of around 200 nm. These polymers, however, are expected to form useful resist materials for the EB and EUV lithography for forming patterns of smaller size than the processing limit of ArF excimer laser because they offer high etching resistance.

[0005] One of the important applications of chemically amplified resist material resides in processing of photomask blanks. Some photomask blanks have a surface material that can have an impact on the pattern profile of the overlying chemically amplified resist film, for example, a layer of a chromium compound, typically chromium oxide deposited on a photomask substrate. For high resolution and profile retention after etching, it is one important performance factor to maintain the profile of a resist film pattern rectangular independent of the type of substrate. In recent years, the multibeam mask writing (MBMW) process is used in the processing of mask blanks to achieve further miniaturization. The resist used in the MBMW process is a low-sensitivity resist (or high-dose region) which is advantageous in roughness while a spotlight is brought to the optimization of the resist composition in the high-dose region.

[0006] Resist compositions for photolithography include positive ones in which exposed areas are dissolved away and negative ones in which exposed areas are left as a pattern. A viable composition is selected among them depending on the desired resist pattern. In general, the chemically amplified negative resist composition comprises a polymer which is normally soluble in an aqueous alkaline developer, an acid generator which is decomposed to generate an acid upon exposure to light, and a crosslinker which causes the polymer to crosslink in the presence of the acid serving as a catalyst, thus rendering the polymer insoluble in the developer (sometimes, the crosslinker is incorporated in the polymer). Most often a quencher is added for controlling the diffusion of the acid generated upon light exposure.

[0007] Typical of the alkali-soluble units to constitute polymers which dissolve in aqueous alkaline developer are units derived from phenols. A number of negative resist compositions of such type were developed, especially as adapted for exposure to KrF excimer laser light. These compositions

have not been used in the ArF excimer laser lithography because the phenolic units are not transmissive to exposure light having a wavelength of 150 to 220 nm. Recently, these compositions are recognized attractive again as the negative resist composition for the short wavelength (e.g., EB or EUV) lithography capable of forming smaller size patterns. Exemplary compositions are described in Patent Documents 1 to 3.

[0008] Of many acid generators known in the art, Patent Document 4 describes a sulfonium salt capable of generating a sulfonic acid having an iodized aromatic group. This sulfonium salt aims to enhance the sensitization effect in the EUV lithography and is mainly handled as a quencher for fluorinated alkane sulfonic acids. The sulfonium salt has not been fully studied as the acid generator, especially for use in resist compositions comprising polyhydroxystyrene as a base polymer as used in the EB writing process in the processing of mask blanks.

[0009] Attempts were made to ameliorate resist sensitivity and pattern profile in a controlled way by properly selecting and combining components used in resist compositions and adjusting processing conditions. One outstanding problem is the diffusion of acid because acid diffusion has a significant impact on the resolution of a chemically amplified resist composition.

[0010] The quencher is, in fact, essential for controlling acid diffusion and improving resist performance, especially resolution. Studies have been made on the quencher while amines and weak acid onium salts have been generally used. The weak acid onium salts are exemplified in several patent documents. For example, Patent Document 5 describes that the addition of triphenylsulfonium acetate ensures to form a satisfactory resist pattern without T-top profile, a difference in line width between isolated and grouped patterns, and standing waves. Patent Document 6 describes the addition of ammonium salts of sulfonic acids or carboxylic acids for achieving improvements in sensitivity, resolution and exposure margin. Also, Patent Document 7 describes that a resist composition for KrF or EB lithography comprising a PAG capable of generating a fluorinated carboxylic acid is improved in resolution and process latitudes such as exposure margin and depth of focus. Patent Document 8 describes that a resist composition for F.sub.2 lithography using F.sub.2 laser comprising a PAG capable of generating a fluorinated carboxylic acid is improved in LER and overcomes the footing problem. These compositions are used in the KrF, EB and F.sub.2 lithography processes.

[0011] Patent Document 9 describes a positive photosensitive composition for ArF lithography comprising a carboxylic acid onium salt. This system is based on the mechanism that a salt exchange occurs between a weak acid onium salt and a strong acid (sulfonic acid) generated by a PAG upon exposure, to form a weak acid and a strong acid onium salt. That is, the strong acid (sulfonic acid) having high acidity is replaced by a weak acid (carboxylic acid), thereby suppressing acid-catalyzed decomposition reaction of acid labile group and reducing or controlling the distance of acid diffusion. The onium salt apparently functions as a quencher.

[0012] In addition to further improvements in roughness, resist compositions are recently demanded which are capable of forming not only line-and-space (LS), isolated line (IL) and isolated space (IS) patterns of satisfactory profile, but also dot patterns of satisfactory profile. Patent Document 10 describes an acid generator capable of generating a bulky acid with controlled diffusion, from which patterns having satisfactory resolution and roughness are obtainable, but the formation of dot patterns is accompanied with corner rounding.

CITATION LIST

[0013] Patent Document 1: JP-A 2006-201532 (US20060166133, EP 1684118) [0014] Patent Document 2: JP-A 2006-215180 [0015] Patent Document 3: JP-A 2008-249762 (U.S. Pat. No. 9,075,306, EP 1975711) [0016] Patent Document 4: JP 6645464 [0017] Patent Document 5: JP 3955384 (U.S. Pat. No. 6,479,210) [0018] Patent Document 6: JP-A H11-327143 [0019] Patent Document 7: JP 4231622 (U.S. Pat. No. 6,485,883) [0020] Patent Document 8: JP 4116340 (U.S. Pat. No. 7,214,467) [0021] Patent Document 9: JP 4226803 (U.S. Pat. No. 6,492,091) [0022] Patent Document 10: JP 6248882

SUMMARY OF THE INVENTION

[0023] An object of the invention is to provide a chemically amplified negative resist composition which exhibits an improved resolution upon pattern formation and forms a pattern with reduced LER and high fidelity, and a resist pattern forming process using the same.

[0024] The inventor has found that when an onium salt containing a sulfonic acid anion having a substituted fused ring structure and an aromatic sulfonic acid structure is added as an acid generator to a resist composition, the onium salt generates an acid having an appropriate acidity and the bulky anion structure is effective for restraining excessive diffusion of the acid. A pattern with minimal LER is obtainable from the resist composition. A dot pattern of satisfactory rectangularity is obtainable by virtue of properly inhibited dissolution.

[0025] In one aspect, the invention provides a chemically amplified negative resist composition comprising (A) a photoacid generator in the form of an onium salt having the formula (A) and (B) a base polymer containing a polymer comprising repeat units having the formula (B1).

##STR00001## [0026] Herein R.sup.1 to R.sup.12 are each independently hydrogen, halogen, or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, [0027] one of R.sup.13 and R.sup.14 is a group having a partial structure having the formula (a) and the other is hydrogen, halogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, [0028] at least two of R.sup.1 to R.sup.14 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and [0029] Z.sup.+ is an onium cation.

##STR00002## [0030] Herein m₁ is 0 or 1, m₂ is an integer of 0 to 4 when m₁=0 and an integer of 0 to 6 when m₁=1, m₃ is an integer of 0 to 3 when m₁=0 and an integer of 0 to 5 when m₁=1, m₂+m₃ is from 0 to 4 when m₁=0 and from 0 to 6 when m₁=1, [0031] R.sup.F is fluorine, a C.sub.1-C.sub.6 fluorinated alkyl group, C.sub.1-C.sub.6 fluorinated alkoxy group, or C.sub.1-C.sub.6 fluorinated alkylthio group, [0032] R.sup.15 is a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, a plurality of R.sup.15 may bond together to form a ring with the carbon atoms to which they are attached when m₃ is 2 or more, [0033] L.sup.A and L.sup.B are each independently a single bond, ether bond, ester bond, amide bond, sulfonate ester bond, sulfonamide bond, carbonate bond or carbamate bond, and [0034] X.sup.L is a single bond or a C.sub.1-C.sub.40 hydrocarbylene group which may contain a heteroatom.

##STR00003## [0035] Herein a₁ is 0 or 1, a₂ is 0, 1 or 2, a₃ is an integer meeting 0≤a₃≤5+2(a₂)-a₄, a₄ is 1, 2 or 3, [0036] R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0037] R.sup.21 is halogen, an optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyl group, optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyloxy group, or optionally halogenated C.sub.2-C.sub.8 saturated hydrocarbylcarbonyloxy group, and [0038] A.sup.1 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group in which some-CH.sub.2— may be replaced by —O—.

[0039] In one preferred embodiment, the onium salt has the formula (A.sup.1):

##STR00004## [0040] wherein R.sup.1 to R.sup.13, R.sup.15, L.sup.A, X.sup.L, R.sup.F, m₁ to m₃, and Z.sup.+ are as defined above.

[0041] In a more preferred embodiment, the onium salt has the formula (A.sup.2):

##STR00005## [0042] wherein R.sup.5, R.sup.10 to R.sup.13, R.sup.15, L.sup.A, X.sup.L, R.sup.F, m₁ to m₃, and Z.sup.+ are as defined above, [0043] m₄ and m₅ are each independently an integer of 0 to 4, [0044] R.sup.16 and R.sup.17 are each independently hydrogen, halogen, or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom; when m₄ is 2 or more, a plurality of R.sup.16 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and when m₅ is 2 or more, a plurality of R.sup.17 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon.

[0045] In one preferred embodiment, Z.sup.+ is a sulfonium cation having the formula (cation-1) or

iodonium cation having the formula (cation-2):

##STR00006## [0046] wherein R.sup.ct1 to R.sup.ct5 are each independently halogen or a C.sub.1-C.sub.30 hydrocarbyl group which may contain a heteroatom, R.sup.ct1 and R.sup.ct2 may bond together to form a ring with the sulfur atom to which they are attached.

[0047] In one preferred embodiment, the polymer further comprises repeat units of at least one type selected from repeat units having the formula (B2) and repeat units having the formula (B3).

##STR00007## [0048] Herein b1 is 0 or 1, b2 is 0, 1 or 2, b3 is an integer meeting $0 \leq b3 \leq 5+2(b2)-b4$, b4 is 1, 2 or 3, b5 is 0, 1 or 2, b6 is 1 or 2, [0049] R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0050] R.sup.31 is halogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, [0051] R.sup.32 and R.sup.33 are each independently hydrogen, a C.sub.1-C.sub.15 saturated hydrocarbyl group which may be substituted with a hydroxy moiety or C.sub.1-C.sub.6 saturated hydrocarbyloxy moiety, or an optionally substituted C.sub.6-C.sub.15 aryl group, excluding that R.sup.32 and R.sup.33 are hydrogen at the same time, R.sup.32 and R.sup.33 may bond together to form a ring with the carbon atom to which they are attached, some —CH.sub.2— in the ring may be replaced by —O— or —S—, [0052] R.sup.34 is halogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, [0053] R.sup.35 and R.sup.36 are each independently hydrogen, a C.sub.1-C.sub.15 saturated hydrocarbyl group which may be substituted with a hydroxy moiety or C.sub.1-C.sub.6 saturated hydrocarbyloxy moiety, or an optionally substituted C.sub.6-C.sub.15 aryl group, excluding that R.sup.35 and R.sup.36 are hydrogen at the same time, R.sup.35 and R.sup.36 may bond together to form a ring with the carbon atom to which they are attached, some —CH.sub.2— in the ring may be replaced by —O— or —S—, [0054] A.sup.2 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group in which some —CH.sub.2— may be replaced by —O—, [0055] W.sup.1 and W.sup.2 are each independently hydrogen, a C.sub.1-C.sub.10 aliphatic hydrocarbyl group, C.sub.2-C.sub.10 aliphatic hydrocarbylcarbonyl group, or optionally substituted C.sub.6-C.sub.15 aryl group.

[0056] In one preferred embodiment, the polymer further comprises repeat units of at least one type selected from repeat units having the formula (B4), repeat units having the formula (B5), and repeat units having the formula (B6).

##STR00008## [0057] Herein c and d are each independently an integer of 0 to 4, e1 is 0 or 1, e2 is 0, 1 or 2, e3 is an integer of 0 to 5, [0058] R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0059] R.sup.41 and R.sup.42 are each independently hydroxy, halogen, an optionally halogenated C.sub.1-C.sub.8 saturated hydrocarbyl group, optionally halogenated C.sub.1-C.sub.8 saturated hydrocarbyloxy group, or optionally halogenated C.sub.2-C.sub.8 saturated hydrocarbylcarbonyloxy group, [0060] R.sup.43 is a C.sub.1-C.sub.20 saturated hydrocarbyl group, C.sub.1-C.sub.20 saturated hydrocarbyloxy group, C.sub.2-C.sub.20 saturated hydrocarbylcarbonyloxy group, C.sub.2-C.sub.20 saturated hydrocarbyloxyhydrocarbyl group, C.sub.2-C.sub.20 saturated hydrocarbylthiohydrocarbyl group, halogen, nitro, cyano, C.sub.1-C.sub.20 saturated hydrocarbylsulfinyl group, or C.sub.1-C.sub.20 saturated hydrocarbylsulfonyl group, and [0061] A.sup.4 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group in which some —CH.sub.2— may be replaced by —O—.

[0062] In one preferred embodiment, the polymer further comprises repeat units of at least one type selected from repeat units having the formulae (B7) to (B14).

##STR00009## ##STR00010## [0063] Herein R.sup.B is each independently hydrogen or methyl, [0064] Y.sup.1 is each independently a single bond, a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, naphthylene group or C.sub.7-C.sub.18 group obtained by combining the foregoing, *-O—Y.sup.11—, *-C(=O)—O—Y.sup.11—, or *-C(=O)—NH—Y.sup.11—, Y.sup.11 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, naphthylene group or C.sub.7-C.sub.18 group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, [0065] Y.sup.2 is each independently a single bond or *-Y.sup.21—C(=O)—O—, Y.sup.21 is a C.sub.1-C.sub.20 hydrocarbylene group which

may contain a heteroatom, [0066] Y^{sup.3} is each independently a single bond, methylene, ethylene, phenylene, fluorinated phenylene, trifluoromethyl-substituted phenylene, *-O—Y^{sup.31}—, *-C(=O)—O—Y^{sup.31}—, or *-C(=O)—NH—Y^{sup.31}—, Y^{sup.31} is a C_{sub.1}-C_{sub.6} aliphatic hydrocarbylene group, phenylene group, fluorinated phenylene group, trifluoromethyl-substituted phenylene group, or C_{sub.7}-C_{sub.20} group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, [0067] designates a point of attachment to the carbon atom in the backbone, [0068] designates a point of attachment to the oxygen atom in the formula, [0069] Y^{sup.4} is each independently a single bond or C_{sub.1}-C_{sub.30} hydrocarbylene group which may contain a heteroatom, [0070] f₁ and f₂ are each independently 0 or 1, f₁ and f₂ are 0 when Y^{sup.4} is a single bond, [0071] R^{sup.51} to R^{sup.68} are each independently halogen or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, R^{sup.51} and R^{sup.52} may bond together to form a ring with the sulfur atom to which they are attached, R^{sup.53} and R^{sup.54}, R^{sup.56} and R^{sup.57}, or R^{sup.59} and R^{sup.60} may bond together to form a ring with the sulfur atom to which they are attached, [0072] R^{sup.HF} is hydrogen or trifluoromethyl, and [0073] Xa^{sup.-} is a non-nucleophilic counter ion.

[0074] In one preferred embodiment, the polymer comprises repeat units having the formula (B1-1), repeat units having the formula (B2-1), (B2-2) or (B3-1), and repeat units having the formula (B8):

##STR00011## ##STR00012## [0075] wherein a₄, b₄, b₆, R^{sup.A}, R^{sup.B}, Y^{sup.2}, R^{sup.32}, R^{sup.33}, R^{sup.35}, R^{sup.36}, R^{sup.53}, R^{sup.54}, R^{sup.55}, and R^{sup.HF} are as defined above.

[0076] In another preferred embodiment, the base polymer (B) further contains a polymer comprising repeat units having formula (B1) and repeat units having formula (B2) or (B3), but not repeat units having formulae (B7) to (B14).

[0077] In one preferred embodiment, repeat units having an aromatic ring structure account for at least 60 mol % of the overall repeat units of the polymer in the base polymer.

[0078] The negative resist composition may further comprise (C) a crosslinker.

[0079] In another embodiment, the negative resist composition is free of a crosslinker.

[0080] The negative resist composition may further comprise (D) a fluorinated polymer comprising repeat units of at least one type selected from repeat units having the formula (D1), repeat units having the formula (D2), repeat units having the formula (D3) and repeat units having the formula (D4) and optionally repeat units of at least one type selected from repeat units having the formula (D5) and repeat units having the formula (D6).

##STR00013## [0081] Herein x is an integer of 1 to 3, y is an integer satisfying 0 ≤ y ≤ 5 + 2z - x, z is 0 or 1, g is an integer of 1 to 3, [0082] R^{sup.C} is each independently hydrogen, fluorine, methyl or trifluoromethyl, [0083] R^{sup.D} is each independently hydrogen or methyl, [0084] R^{sup.101}, R^{sup.102}, R^{sup.104} and R^{sup.105} are each independently hydrogen or a C_{sub.1}-C_{sub.10} saturated hydrocarbyl group, [0085] R^{sup.103}, R^{sup.106}, R^{sup.107} and R^{sup.108} are each independently hydrogen, a C_{sub.1}-C_{sub.15} hydrocarbyl group, C_{sub.1}-C_{sub.15} fluorinated hydrocarbyl group, or acid labile group, and when R^{sup.103}, R^{sup.106}, R^{sup.107} and R^{sup.108} each are a hydrocarbyl or fluorinated hydrocarbyl group, an ether bond or carbonyl moiety may intervene in a carbon-carbon bond, [0086] R^{sup.109} is hydrogen or a C_{sub.1}-C_{sub.5} straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond, [0087] R^{sup.110} is a C_{sub.1}-C_{sub.5} straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond, [0088] R^{sup.111} is a C_{sub.1}-C_{sub.20} saturated hydrocarbyl group in which at least one hydrogen is substituted by fluorine, and in which some-CH_{sub.2}— may be replaced by an ester bond or ether bond, [0089] Z^{sup.1} is a C_{sub.1}-C_{sub.20} (g+1)-valent hydrocarbon group or C_{sub.1}-C_{sub.20} (g+1)-valent fluorinated hydrocarbon group, [0090] Z^{sup.2} is a single bond, *-C(=O)—O— or *-C(=O)—NH—, * designates a point of attachment to the carbon atom in the backbone, [0091] Z^{sup.3} is a single bond, —O—, *-C(=O)—O—Z^{sup.31}—Z^{sup.32}— or *-C(=O)—NH—Z^{sup.31}—

Z.sup.32—, Z.sup.31 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group, Z.sup.32 is a single bond, ester bond, ether bond, or sulfonamide bond, and * designates a point of attachment to the carbon atom in the backbone.

[0092] The negative resist composition may further comprise (E) a quencher. Preferably, the acid generator (A) and the quencher (E) are present in a weight ratio of less than 6/1.

[0093] The negative resist composition may further comprise (E) an organic solvent.

[0094] In another aspect, the invention provides a resist pattern forming process comprising the steps of: [0095] applying the chemically amplified negative resist composition defined herein onto a substrate to form a resist film thereon, [0096] exposing the resist film patternwise to high-energy radiation, and [0097] developing the exposed resist film in an alkaline developer.

[0098] Typically, the high-energy radiation is EUV of wavelength 3 to 15 nm or EB.

[0099] In one embodiment, the substrate has the outermost surface of a material containing at least one element selected from chromium, silicon, tantalum, molybdenum, cobalt, nickel, tungsten, and tin.

[0100] The substrate is typically a mask blank of transmission or reflection type.

[0101] Also contemplated herein is a mask blank of transmission or reflection type which is coated with the chemically amplified negative resist composition defined herein.

Advantageous Effects of Invention

[0102] Owing to the onium salt having formula (A), the chemically amplified negative resist composition of the invention is effective for controlling acid diffusion during the exposure step.

When the composition is coated as a resist film and processed to form a pattern, the resist film exhibits a very high resolution during pattern formation and forms a pattern with high fidelity and reduced LER. Owing to the repeat units having formula (B1), when the resist composition is coated onto a substrate to form a resist film, the adhesion of the resist film to the substrate is improved. Also, the dissolution of the resist film in alkaline developer is controlled.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0103] The only FIGURE, FIG. 1 is a diagram showing the ¹H-NMR/DMSO-d₆ spectrum of onium salt PAG-1 synthesized in Synthesis Example 1-1.

DETAILED DESCRIPTION OF THE INVENTION

[0104] As used herein, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. “Optional” or “optionally” means that the subsequently described event or circumstances may or may not occur, and that description includes instances where the event or circumstance occurs and instances where it does not. The notation (C_n-C_m) means a group containing from n to m carbon atoms per group. In chemical formulae, Me stands for methyl, Ac stands for acetyl, and the broken line (---) and asterisk (*) designate a point of attachment or valence bond. The terms “group” and “moiety” are interchangeable.

[0105] The abbreviations and acronyms have the following meaning. [0106] EB: electron beam

[0107] EUV: extreme ultraviolet [0108] Mw: weight average molecular weight [0109] Mn: number

average molecular weight [0110] Mw/Mn: molecular weight distribution or dispersity [0111] GPC:

gel permeation chromatography [0112] PEB: post-exposure bake [0113] PAG: photoacid generator

[0114] LER: line edge roughness

[0115] It is understood that for some structures represented by chemical formulae, there can exist enantiomers and diastereomers because of the presence of asymmetric carbon atoms. In such a case, a single formula collectively represents all such isomers. The isomers may be used alone or in admixture.

[Negative Resist Composition]

[0116] One embodiment of the invention is a chemically amplified negative resist composition comprising (A) a photoacid generator in the form of an onium salt containing a sulfonic acid anion having a substituted fused ring structure and an aromatic sulfonic acid structure, and (B) a base polymer containing a specific polymer.

(a) Photoacid Generator

[0117] The photoacid generator as component (A) is an onium salt having the formula (A).

##STR00014##

[0118] In formula (A), R^{sup.1} to R^{sup.12} are each independently hydrogen, halogen, or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom. One of R^{sup.13} and R^{sup.14} is a group having a partial structure having the formula (a) shown later while the other is hydrogen, halogen or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom.

[0119] Suitable halogen atoms include fluorine, chlorine, bromine, and iodine. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C_{sub.1}-C_{sub.20} alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, n-hexyl, n-octyl, n-nonyl, n-decyl, undecyl, dodecyl, tridecyl, tetradecyl, pentadecyl, heptadecyl, octadecyl, nonadecyl, and icosyl; C_{sub.3}-C_{sub.20} cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclopentyl, cyclohexyl, cyclopropylmethyl, 4-methylcyclohexyl, cyclohexylmethyl, norbornyl, and adamantyl; C_{sub.2}-C_{sub.20} alkenyl groups such as vinyl, allyl, propenyl, butenyl, and hexenyl; C_{sub.3}-C_{sub.20} cyclic unsaturated hydrocarbyl groups such as cyclohexenyl; C_{sub.6}-C_{sub.20} aryl groups such as phenyl and naphthyl; C_{sub.7}-C_{sub.20} aralkyl groups such as benzyl, 1-phenylethyl, and 2-phenylethyl, and combinations thereof. Inter alia, aryl groups are preferred. In the hydrocarbyl group, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some constituent —CH_{sub.2}— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, cyano, fluorine, chlorine, bromine, iodine, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety.

[0120] At least two of R^{sup.1} to R^{sup.14} may bond together to form a ring with the carbon to which they are attached, or the carbon to which they are attached and the intervenient carbon. Examples of the ring thus formed include aliphatic rings such as cyclopropane, cyclobutane, cyclopentane, cyclohexane, norbornane, and adamantane rings and aromatic rings such as benzene, naphthalene and anthracene rings. In the ring, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some constituent —CH_{sub.2}— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the ring may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety. Of the rings, aromatic rings are preferred, and benzene rings are more preferred.

[0121] In formula (A), one of R^{sup.13} and R^{sup.14} is a group having a partial structure having the following formula (a). It is preferred from the aspect of synthesis that R^{sup.14} be a group having a partial structure having the formula (a).

##STR00015##

[0122] In formula (a), m₁ is 0 or 1. The relevant structure is a benzene ring when m₁=0, and a naphthalene ring when m₁=1. The benzene ring corresponding to m₁=0 is preferred from the aspect of solvent solubility. The subscript m₂ is an integer of 0 to 4 when m₁=0 and an integer of 0 to 6 when m₁=1. Preferably, m₂ is 4 when m₁=0, and m₂ is 4, 5 or 6 when m₁=1. The subscript m₃ is an integer of 0 to 3 when m₁=0 and an integer of 0 to 5 when m₁=1. The sum m₂+m₃ is an integer of 0 to 4 when m₁=0 and an integer of 0 to 6 when m₁=1.

[0123] In formula (a), R^{sup.F} is fluorine, a C_{sub.1}-C_{sub.6} fluorinated alkyl group, C_{sub.1}-

C.sub.6 fluorinated alkoxy group, or C.sub.1-C.sub.6 fluorinated alkylthio group. R.sup.F is preferably fluorine, trifluoromethyl, difluoromethyl, trifluoromethoxy, difluoromethoxy, trifluoromethylthio or difluoromethylthio, more preferably fluorine, trifluoromethyl or trifluoromethoxy. When fluorine or a fluorinated substituent group is contained, the acid strength of the generated acid is so enhanced due to the electron attractive effect that deprotection reaction of acid labile groups such as tertiary ester or tertiary ether groups may smoothly take place. When m₂ is 2, 3 or 4, a plurality of R.sup.F may be identical or different.

[0124] In formula (a), R.sup.15 is a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for the hydrocarbyl group R.sup.1, but not limited thereto. When m₃ is 2 or more, a plurality of R.sup.15 may be identical or different. When m₃ is 2 or more, a plurality of R.sup.15 may bond together to form a ring with the carbon atoms to which they are attached. The ring is preferably a 5 to 8-membered ring.

[0125] In formula (a), L.sup.A and L.sup.B are each independently a single bond, ether bond, ester bond, amide bond, sulfonate ester bond, sulfonamide bond, carbonate bond or carbamate bond. Inter alia, a single bond, ether bond or ester bond is preferred.

[0126] In formula (a), X.sup.L is a single bond or a C.sub.1-C.sub.40 hydrocarbylene group which may contain a heteroatom. The hydrocarbylene group may be straight, branched or cyclic and examples thereof include alkanediyl groups, cyclic saturated hydrocarbylene groups, and arylene groups. Suitable heteroatoms include oxygen, nitrogen and sulfur.

[0127] Examples of the C.sub.1-C.sub.40 hydrocarbylene group which may contain a heteroatom, represented by X.sup.L, are shown below, but not limited thereto. Herein, * designates a point of attachment to L.sup.A or L.sup.B.

##STR00016## ##STR00017## ##STR00018## ##STR00019## ##STR00020## ##STR00021##

[0128] Of these, X.sup.L-0 to X.sup.L-22 and X.sup.L-47 to X.sup.L-58 are preferred.

[0129] Of the onium salts having formula (A), those having the formula (A.sup.1) are preferred.

##STR00022##

[0130] Herein R.sup.1 to R.sup.13, R.sup.15, L.sup.A, X.sup.L, R.sup.F, m₁ to m₃, and Z.sup.+ are as defined above.

[0131] Of the onium salts having formula (A.sup.1), those having the formula (A.sup.2) are more preferred.

##STR00023##

[0132] In formula (A.sup.2), R.sup.5, R.sup.10 to R.sup.13, R.sup.15, L.sup.A, X.sup.L, R.sup.F, m₁ to m₃, and Z.sup.+ are as defined above.

[0133] In formula (A.sup.2), m₄ and m₅ are each independently an integer of 0 to 4. From the aspect of reactant availability, m₄ and m₅ each are preferably 0, 1 or 2.

[0134] In formula (A.sup.2), R.sup.16 and R.sup.17 are each independently hydrogen, halogen, or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. Examples of the halogen and hydrocarbyl group are as exemplified above for the halogen and hydrocarbyl group represented by R.sup.1 to R.sup.14.

[0135] When m₄ is 2 or more, a plurality of R.sup.16 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon. When m₅ is 2 or more, a plurality of R.sup.17 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon. Examples of the ring are as exemplified above for the ring that is formed by at least two of R.sup.1 to R.sup.14.

[0136] Examples of the anion in the onium salt having formula (A) are shown below, but not limited thereto. It is noted that structures obtained by combining functional groups in different exemplary structures are also included herein.

##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028## ##STR00029##

[illegible]

##STR00310## ##STR00311## ##STR00312## ##STR00313## ##STR00314## ##STR00315##
##STR00316## ##STR00317##
##STR00318## ##STR00319## ##STR00320## ##STR00321## ##STR00322## ##STR00323##
##STR00324## ##STR00325## ##STR00326## ##STR00327## ##STR00328## ##STR00329##
##STR00330## ##STR00331## ##STR00332## ##STR00333## ##STR00334## ##STR00335##
##STR00336## ##STR00337## ##STR00338## ##STR00339## ##STR00340## ##STR00341##
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##STR00360## ##STR00361## ##STR00362## ##STR00363## ##STR00364## ##STR00365##
##STR00366## ##STR00367## ##STR00368## ##STR00369## ##STR00370## ##STR00371##
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##STR00390##

##STR00391## ##STR00392## ##STR00393## ##STR00394## ##STR00395## ##STR00396##
##STR00397## ##STR00398## ##STR00399## ##STR00400## ##STR00401## ##STR00402##
##STR00403## ##STR00404## ##STR00405## ##STR00406## ##STR00407## ##STR00408##
##STR00409## ##STR00410## ##STR00411## ##STR00412## ##STR00413## ##STR00414##
##STR00415## ##STR00416## ##STR00417## ##STR00418## ##STR00419## ##STR00420##
##STR00421## ##STR00422## ##STR00423## ##STR00424## ##STR00425## ##STR00426##
##STR00427## ##STR00428## ##STR00429## ##STR00430## ##STR00431## ##STR00432##
##STR00433## ##STR00434## ##STR00435## ##STR00436## ##STR00437## ##STR00438##
##STR00439## ##STR00440## ##STR00441## ##STR00442## ##STR00443## ##STR00444##
##STR00445## ##STR00446## ##STR00447## ##STR00448##

[0137] In formula (A), Z^{sup.}+ is an onium cation, preferably sulfonium cation having the formula (cation-1) or iodonium cation having the formula (cation-2).

##STR00449##

[0138] In formulae (cation-1) and (cation-2), R^{sup.ct1} to R^{sup.ct5} are each independently halogen or a C_{sub.1}-C_{sub.30} hydrocarbyl group which may contain a heteroatom. Suitable halogen atoms include fluorine, chlorine, bromine and iodine. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C_{sub.1}-C_{sub.30} alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl; C_{sub.3}-C_{sub.30} cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclopentyl, cyclohexyl, cyclopropylmethyl, 4-methylcyclohexyl, cyclohexylmethyl, norbornyl, adamantyl; C_{sub.2}-C_{sub.30} alkenyl groups such as vinyl, allyl, propenyl, butenyl, hexenyl; C_{sub.3}-C_{sub.30} cyclic unsaturated hydrocarbyl groups such as cyclohexenyl; C_{sub.6}-C_{sub.30} aryl groups such as phenyl, naphthyl, thienyl; C_{sub.7}-C_{sub.30} aralkyl groups such as benzyl, 1-phenylethyl, 2-phenylethyl, and combinations thereof. Inter alia, the aryl groups are preferred. In the hydrocarbyl groups, some or all hydrogen may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some-CH_{sub.2}— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, nitro, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety.

[0139] Also, R^{sup.ct1} and R^{sup.ct2} may bond together to form a ring with the sulfur atom to which they are attached. Exemplary structures of the ring are shown below.

##STR00450##

[0140] The broken line designates a point of attachment to R^{sup.ct3}.

[0141] Examples of the sulfonium cation having formula (cation-1) are shown below, but not

limited thereto.

##STR00451## ##STR00452## ##STR00453## ##STR00454## ##STR00455## ##STR00456##
##STR00457## ##STR00458## ##STR00459## ##STR00460## ##STR00461## ##STR00462##
##STR00463## ##STR00464## ##STR00465## ##STR00466## ##STR00467## ##STR00468##
##STR00469## ##STR00470## ##STR00471## ##STR00472## ##STR00473##
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##STR00486## ##STR00487## ##STR00488## ##STR00489## ##STR00490## ##STR00491##
##STR00492## ##STR00493## ##STR00494##

##STR00495## ##STR00496## ##STR00497## ##STR00498## ##STR00499## ##STR00500##
##STR00501## ##STR00502## ##STR00503## ##STR00504## ##STR00505## ##STR00506##
##STR00507## ##STR00508## ##STR00509## ##STR00510## ##STR00511## ##STR00512##

[0142] Examples of the iodonium cation having formula (cation-2) are shown below, but not limited thereto.

##STR00513## ##STR00514## ##STR00515##

[0143] Examples of the onium salt include arbitrary combinations of anions with cations, both as exemplified above.

[0144] The onium salt can be synthesized by well-known methods. For example, a method of preparing an onium salt having the formula (PAG-1-ex) is described.

##STR00516##

[0145] Herein R.sup.1 to R.sup.13, R.sup.F, R.sup.15, L.sup.A, X.sup.L, m1 to m3, and Z.sup.+ are as defined above, M.sup.+ is a counter cation, and X.sup.- is a counter anion.

[0146] The first step is to react reactant SM-1 with reactant SM-2, which are commercially available or can be synthesized by any well-known synthesis method, to form an Intermediate In-1. A variety of condensing agents may be used when an ester bond is directly formed from the carboxy group on reactant SM-1 and the hydroxy group on reactant SM-2. Examples of the condensing agent used herein include N,N'-dicyclohexylcarbodiimide, N,N'-diisopropylcarbodiimide, 1-[3-(dimethylamino) propyl]-3-ethylcarbodiimide, and 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride. From the standpoint of ease of removal of urea compounds formed as by-products after reaction, it is desirable to use 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride. The reaction is carried out by dissolving reactants SM-1 and SM-2 in a halogenated solvent such as methylene chloride and adding a condensing agent. The reaction rate can be increased by adding 4-dimethylaminopyridine (DMAP) as a catalyst. While it is desirable in view of yield to monitor the reaction by silica gel thin layer chromatography (TLC) until the reaction is complete, the reaction time is typically about 12 to 24 hours. After the reaction is terminated, urea compounds formed by side reactions are removed by filtration or water-washing if necessary. The reaction solution is subjected to standard aqueous work-up, obtaining Intermediate In-1. If necessary, the intermediate is purified by a standard technique such as chromatography or recrystallization.

[0147] The second step is a salt exchange between Intermediate In-1 and reactant SM-3 in the form of an onium salt: Z.sup.+X.sup.- to synthesize an onium salt PAG-1-ex. X.sup.- is preferably a chloride ion, bromide ion, iodide ion or methylsulfate anion because the exchange reaction readily takes place in a quantitative way. It is desirable in view of yield to monitor the reaction process by TLC. The reaction mixture is subjected to standard aqueous work-up, obtaining an onium salt PAG-1-ex. If necessary, the onium salt is purified by a standard technique such as chromatography or recrystallization.

[0148] In the above-illustrated scheme, the second step of ion exchange may be readily carried out by any well-known procedure, for example, with reference to JP-A 2007-145797.

[0149] It is noted that the preparation method according to the above scheme is merely exemplary and the method of preparing the inventive onium salt is not limited thereto.

[0150] The inventive onium salt is structurally characterized in that a substituted fused ring structure and an aromatic sulfonic acid structure are introduced in the anion. The substituted fused ring structure has a large excluded volume and thus serves as a bulky substituent to effectively inhibit the generated acid from diffusing. The substituted fused ring structure is also enough resistant to alkaline developer to prevent the pattern in unexposed region from losing a film thickness. The aromatic sulfonic acid structure, on the other hand, generates an acid of robust structure, which is effective for inhibiting acid diffusion. It is preferred that the aromatic ring that forms the aromatic sulfonic acid structure contain fluorine or an electron attractive sulfonate ester bond as a linking group. Then the generated acid has a higher acidity enough to efficiently deprotect the acid labile groups in the base polymer. Since fluorine has a high EUV absorption effect, which is not so high as iodine, the inclusion of more fluorine atoms leads to a greater number of secondary electrons, which promotes decomposition of cations and contributes to a sensitivity increase. JP 7109178 proposes an alkanesulfonic acid type photoacid generator having 2 to 4 fluorine atoms, which has the problems of noticeable acid diffusion because of alkanesulfonic acid and poor solvent solubility which leaves the risk of development defects. By virtue of the synergy of these effects, the resist composition comprising the onium salt can form a resist pattern having a high sensitivity, low acid diffusion, and improved LWR of line patterns or improved CDU of hole patterns. The pattern is fully resistant to collapse. The resist composition is effective for forming small-size patterns.

[0151] In the chemically amplified negative resist composition, the amount of the photoacid generator (A) is preferably 0.1 to 40 parts by weight, more preferably 1 to 20 parts by weight per 80 parts by weight of a base polymer to be described just below. As long as the amount of photoacid generator (A) is in the range, an acid is generated in an amount necessary for deprotection of acid labile groups so that a pattern of satisfactory profile may be formed, and the composition has storage stability. The photoacid generator (A) may be used alone or in admixture of two or more.

[0152] In the resist composition, an acid generator other than the onium salt having formula (A), which is referred to as other acid generator, hereinafter, may be added for the purpose of correcting the profile of patterns. The other acid generator may be selected from well-known acid generators for resist compositions. The amount of the other acid generator is preferably 0 to 40 parts by weight, more preferably 0 to 30 parts by weight per 80 parts by weight of the base polymer (B) in view of sensitivity and acid diffusion-suppressing effect. The other acid generator may be used alone or in admixture.

(B) Base Polymer

[0153] The base polymer as component (B) contains a polymer comprising repeat units having the following formula (B1). Notably, the polymer is referred to as polymer B and the units having formula (B1) are also referred to as units B1. The repeat units B1 are effective for providing etch resistance, adhesion to substrates, and solubility in alkaline developer.

##STR00517##

[0154] In formula (B1), a₁ is 0 or 1. The subscript a₂ is 0, 1 or 2. The corresponding structure represents a benzene skeleton when a₂=0, a naphthalene skeleton when a₂=1, and an anthracene skeleton when a₂=2. The subscript a₃ is an integer in the range: $0 \leq a_3 \leq 5 + 2(a_2) - a_4$, and a₄ is 1, 2 or 3. In case of a₂=0, preferably a₃ is 0, 1, 2 or 3, and a₄ is 1, 2 or 3. In case of a₂=1 or 2, preferably a₃ is 0, 1, 2, 3 or 4, and a₄ is 1, 2 or 3.

[0155] In formula (B1), R^{sup.A} is hydrogen, fluorine, methyl or trifluoromethyl.

[0156] In formula (B1), R^{sup.21} is halogen, an optionally halogenated C_{sub.1}-C_{sub.6} saturated hydrocarbyl group, optionally halogenated C_{sub.1}-C_{sub.6} saturated hydrocarbyloxy group or optionally halogenated C_{sub.2}-C_{sub.8} saturated hydrocarbylcarbonyloxy group. The saturated hydrocarbyl group and saturated hydrocarbyl moiety in the saturated hydrocarbyloxy group and saturated hydrocarbylcarbonyloxy group may be straight, branched or cyclic. Examples thereof

include alkyl groups such as methyl, ethyl, n-propyl, isopropyl, butyl, pentyl, hexyl and structural isomers thereof; cycloalkyl groups such as cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl; and combinations thereof. A carbon count within the upper limit ensures a sufficient solubility in alkaline developer. When a_3 is 2 or more, a plurality of $R_{sup.21}$ may be identical or different.

[0157] In formula (B1), $A_{sup.1}$ is a single bond or a $C_{sub.1}$ - $C_{sub.10}$ saturated hydrocarbylene group in which some constituent $-CH_{sub.2}-$ may be replaced by $-O-$. The saturated hydrocarbylene group may be straight, branched or cyclic and examples thereof include $C_{sub.1}$ - $C_{sub.10}$ alkanediyl groups such as methylene, ethane-1,2-diyl, propane-1,3-diyl, butane-1,4-diyl, pentane-1,5-diyl, hexane-1,6-diyl, and structural isomers thereof; $C_{sub.3}$ - $C_{sub.10}$ cyclic saturated hydrocarbylene groups such as cyclopropanediyl, cyclobutanediyl, cyclopentanedyl, and cyclohexanedyl; and combinations thereof. For the saturated hydrocarbylene group containing an ether bond, in case of $a_1=1$ in formula (B1), the ether bond may be incorporated at any position excluding the position between the α - and β -carbons relative to the ester oxygen. In case of $a_1=0$, the atom bonding to the backbone becomes an ether oxygen atom, and a second ether bond may be incorporated at any position excluding the position between the α - and β -carbons relative to the ether oxygen. Saturated hydrocarbylene groups having no more than 10 carbon atoms are desirable because of a sufficient solubility in alkaline developer.

[0158] Preferred examples of the repeat units B1 wherein $a_1=0$ and $A_{sup.1}$ is a single bond (meaning that the aromatic ring is directly bonded to the main chain of the polymer), that is, repeat units free of a linker: $-C(=O)-O-A_{sup.1}-$ include units derived from 3-hydroxystyrene, 4-hydroxystyrene, 5-hydroxy-2-vinylnaphthalene, and 6-hydroxy-2-vinylnaphthalene. More preferred are repeat units having the formula (B1-1).

##STR00518##

[0159] Herein $R_{sup.A}$ and a_4 are as defined above.

[0160] Preferred examples of the repeat units B1 wherein $a_1=1$, that is, having a linker: $-CO-O-A_{sup.1}-$ are shown below, but not limited thereto. Herein $R_{sup.A}$ is as defined above.

##STR00519##

[0161] The repeat units B1 may be of one type or a combination of plural types.

[0162] The polymer B may further comprise repeat units of at least one type selected from repeat units having the formula (B2) and repeat units having the formula (B3). Notably the repeat units having formulae (B2) and (B3) are simply referred to as repeat units B2 and B3. Of the polymers B, a polymer further comprising repeat units B2 and/or B3 is referred to as polymer B'.

##STR00520##

[0163] Upon exposure to high-energy radiation, the repeat unit B2 or B3 functions such that $-O-W_{sup.1}$ or $-O-W_{sup.2}$ undergoes elimination reaction under the action of an acid which is generated by the acid generator. That is, the unit B2 or B3 induces insolubilization in alkaline developer and crosslinking reaction between polymer molecules. The repeat unit B2 or B3 provides for efficient progress of negative-working reaction, leading to an improvement in resolution performance.

[0164] In formula (B2), b_1 is 0 or 1, b_2 is 0, 1 or 2, b_3 is an integer in the range: $0 \leq b_3 \leq 5+2(b_2)-b_4$, and b_4 is 1, 2 or 3.

[0165] In formula (B3), b_5 is 0, 1 or 2, preferably 0 or 1, and b_6 is 1 or 2, preferably 1.

[0166] In formulae (B2) and (B3), $R_{sup.A}$ is hydrogen, fluorine, methyl or trifluoromethyl. In formula (B3), $R_{sup.A}$ is preferably hydrogen or methyl, more preferably hydrogen.

[0167] In formula (B2), $R_{sup.31}$ is halogen or a $C_{sub.1}$ - $C_{sub.20}$ hydrocarbyl group which may contain a heteroatom. Suitable halogen atoms include fluorine, chlorine, bromine and iodine. The $C_{sub.1}$ - $C_{sub.20}$ hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic, and examples thereof are as exemplified above for the hydrocarbyl groups $R_{sup.1}$ to $R_{sup.12}$ in formula (A). When b_3 is 2 or more, a plurality of $R_{sup.31}$ may be identical or different.

[0168] In formula (B2), R.sup.32 and R.sup.33 are each independently hydrogen, a C.sub.1-C.sub.15 saturated hydrocarbyl group or a C.sub.6-C.sub.15 aryl group. The hydrocarbyl group may be substituted with a hydroxy moiety or C.sub.1-C.sub.6 saturated hydrocarbyloxy moiety and the aryl group may be substituted. It is excluded that R.sup.32 and R.sup.33 are hydrogen at the same time. R.sup.32 and R.sup.33 may bond together to form a ring with the carbon atom to which they are attached, in which some —CH.sub.2— may be replaced by —O— or —S—. R.sup.32 and R.sup.33 are preferably selected from alkyl groups such as methyl, ethyl, propyl, butyl and structural isomers thereof, and substituted forms of the foregoing alkyl groups in which some hydrogen is substituted by a hydroxy or saturated hydrocarbyloxy moiety.

[0169] In formula (B3), R.sup.34 is halogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. Suitable halogen atoms include fluorine, chlorine, bromine and iodine. The C.sub.1-C.sub.20 hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for the hydrocarbyl groups represented by R.sup.1 to R.sup.12 in formula (A). When b5 is 2 or more, a plurality of R.sup.34 may be identical or different.

[0170] In formula (b3), R.sup.35 and R.sup.36 are each independently hydrogen, a C.sub.1-C.sub.15 saturated hydrocarbyl group or a C.sub.6-C.sub.15 aryl group. The hydrocarbyl group may be substituted with a hydroxy or C.sub.1-C.sub.6 saturated hydrocarbyloxy moiety and the aryl group may be substituted.

[0171] The C.sub.1-C.sub.15 saturated hydrocarbyl groups R.sup.35 and R.sup.36 may be straight, branched or cyclic, and examples thereof are as exemplified above for the hydrocarbyl groups R.sup.32 and R.sup.33.

[0172] Examples of the C.sub.6-C.sub.15 aryl groups R.sup.35 and R.sup.36 include phenyl, naphthyl and anthryl, with phenyl being preferred. The aryl group may have a substituent, examples of which include halogen, optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyl moieties, and optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyloxy moieties.

[0173] It is excluded that R.sup.35 and R.sup.36 are hydrogen at the same time. When one of R.sup.35 and R.sup.36 is an optionally substituted aryl group, the preferred substituent on the other of R.sup.35 and R.sup.36 is hydrogen.

[0174] Preferably R.sup.35 and R.sup.36 are identical, more preferably both methyl.

[0175] R.sup.35 and R.sup.36 may bond together to form a ring with the carbon atom to which they are attached. In the ring, some —CH.sub.2— may be replaced by —O— or —S—. Examples of the ring include cyclopropane, cyclobutene, cyclopentane, cyclohexane, norbornane, adamantane, tricyclo[5.2.1.0.sup.2,6]decane, tetracyclo[6.2.1.1.sup.3,6.0.sup.2,7]dodecane, oxanorbornane, and thianorbornane rings, but are not limited thereto.

[0176] In formula (B2), A.sup.2 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group in which some constituent —CH.sub.2— may be replaced by —O—. The saturated hydrocarbylene group may be straight, branched or cyclic and examples thereof include alkanediyl groups such as methylene, ethane-1,2-diyl, propane-1,3-diyl, butane-1,4-diyl, pentane-1,5-diyl, hexane-1,6-diyl, and structural isomers thereof; cyclic saturated hydrocarbylene groups such as cyclopropanediyl, cyclobutanediyl, cyclopentanediiyl, and cyclohexanediiyl; and combinations thereof. For the saturated hydrocarbylene group containing an ether bond, in case of b1=1 in formula (B2), the ether bond may be incorporated at any position excluding the position between the α - and β -carbons relative to the ester oxygen. In case of b1=0, the atom bonding to the backbone becomes an ether oxygen atom, and a second ether bond may be incorporated at any position excluding the position between the α - and β -carbons relative to the ether oxygen.

[0177] In formulae (B2) and (B3), W.sup.1 and W.sup.2 are each independently hydrogen, a C.sub.1-C.sub.10 aliphatic hydrocarbyl group, C.sub.2-C.sub.10 aliphatic hydrocarbylcarbonyl group, or an optionally substituted C.sub.6-C.sub.15 aryl group.

[0178] The C.sub.1-C.sub.10 aliphatic hydrocarbyl groups W.sup.1 and W.sup.2 may be saturated

or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.10 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, neopentyl, n-hexyl, n-heptyl, n-octyl, 2-ethylhexyl, n-nonyl, and n-decyl; C.sub.3-C.sub.10 cyclic saturated hydrocarbyl groups such as cyclopentyl and cyclohexyl; C.sub.2-C.sub.10 alkenyl groups such as vinyl, 1-propenyl, 2-propenyl, butenyl and hexenyl; and C.sub.3-C.sub.10 cyclic unsaturated hydrocarbyl groups such as cyclohexenyl. The hydrocarbyl moiety in the C.sub.2-C.sub.10 aliphatic hydrocarbylcarbonyl groups W.sup.1 and W.sup.2 may be saturated or unsaturated and straight, branched or cyclic, and examples thereof include those illustrated above for the C.sub.1-C.sub.10 aliphatic hydrocarbyl groups W.sup.1 and W.sup.2, but of 1 to 9 carbon atoms. Examples of the aryl groups W.sup.1 and W.sup.2 include phenyl, naphthyl and anthryl, with phenyl being preferred. The aryl group may have a substituent, examples of which include halogen, optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyl moieties, and optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyloxy moieties.

[0179] Of the repeat units B2, repeat units having formula (B2-1) or (B2-2) are preferred.

##STR00521##

[0180] Herein R.sup.A, R.sup.32, R.sup.33, and b4 are as defined above.

[0181] Preferred examples of the repeat unit B2 are given below, but not limited thereto. Herein R.sup.A is as defined above.

##STR00522## ##STR00523## ##STR00524## ##STR00525## ##STR00526## ##STR00527##
##STR00528## ##STR00529## ##STR00530## ##STR00531## ##STR00532## ##STR00533##
##STR00534## ##STR00535##

##STR00536## ##STR00537## ##STR00538## ##STR00539## ##STR00540## ##STR00541##
##STR00542## ##STR00543## ##STR00544##

[0182] The repeat unit B2 may be of one type or a combination of plural types.

[0183] Of the repeat units B3, repeat units having formula (B3-1) are preferred.

##STR00545##

[0184] Herein R.sup.A, R.sup.35, R.sup.36, and b6 are as defined above.

[0185] Preferred examples of the repeat unit B3 are given below, but not limited thereto. Herein R.sup.A is as defined above.

##STR00546## ##STR00547## ##STR00548## ##STR00549## ##STR00550## ##STR00551##
##STR00552## ##STR00553## ##STR00554## ##STR00555## ##STR00556## ##STR00557##
##STR00558## ##STR00559## ##STR00560## ##STR00561## ##STR00562## ##STR00563##
##STR00564## ##STR00565## ##STR00566## ##STR00567## ##STR00568## ##STR00569##
##STR00570## ##STR00571## ##STR00572## ##STR00573## ##STR00574##
##STR00575## ##STR00576## ##STR00577## ##STR00578## ##STR00579## ##STR00580##
##STR00581## ##STR00582## ##STR00583## ##STR00584## ##STR00585## ##STR00586##
##STR00587## ##STR00588## ##STR00589## ##STR00590## ##STR00591## ##STR00592##
##STR00593## ##STR00594## ##STR00595## ##STR00596## ##STR00597## ##STR00598##
##STR00599## ##STR00600## ##STR00601## ##STR00602## ##STR00603## ##STR00604##
##STR00605## ##STR00606## ##STR00607## ##STR00608## ##STR00609## ##STR00610##
##STR00611## ##STR00612## ##STR00613## ##STR00614## ##STR00615## ##STR00616##
##STR00617##
##STR00618## ##STR00619##

[0186] The repeat unit B3 may be of one type or a combination of plural types.

[0187] For the purpose of improving etch resistance, preferably the polymer B or B' further comprises repeat units of at least one type selected from repeat units having the formula (B4), repeat units having the formula (B5) and repeat units having the formula (B6). Notably these units are simply referred to as repeat units B4, B5 and B6.

##STR00620##

[0188] In formulae (B4) and (B5), c and d are each independently an integer of 0 to 4.

[0189] In formulae (B4) and (B5), R^{sup.41} and R^{sup.42} are each independently hydroxy, halogen, an optionally halogenated C_{sub.1}-C_{sub.8} saturated hydrocarbyl group, optionally halogenated C_{sub.1}-C_{sub.8} saturated hydrocarbyloxy group, or optionally halogenated C_{sub.2}-C_{sub.8} saturated hydrocarbylcarbonyloxy group. The saturated hydrocarbyl group, saturated hydrocarbyloxy group, and saturated hydrocarbylcarbonyloxy group may be straight, branched or cyclic. When c is 2 or more, a plurality of R^{sup.41} may be identical or different. When d is 2 or more, a plurality of R^{sup.42} may be identical or different.

[0190] In formula (B6), e₁ is 0 or 1. The subscript e₂ is 0, 1 or 2; the corresponding structure represents a benzene skeleton when e₂=0, a naphthalene skeleton when e₂=1, and an anthracene skeleton when e₂=2. The subscript e₃ is an integer of 0 to 5. In case of e₂=0, preferably e₃ is an integer of 0 to 3. In case of e₂=1 or 2, preferably e₃ is an integer of 0 to 4.

[0191] In formula (B6), R^{sup.A} is hydrogen, fluorine, methyl or trifluoromethyl.

[0192] In formula (B6), R^{sup.43} is a C_{sub.1}-C_{sub.20} saturated hydrocarbyl group, C_{sub.1}-C_{sub.20} saturated hydrocarbyloxy group, C_{sub.2}-C_{sub.20} saturated hydrocarbylcarbonyloxy group, C_{sub.2}-C_{sub.20} saturated hydrocarbyloxyhydrocarbyl group, C_{sub.2}-C_{sub.20} saturated hydrocarbylthiohydrocarbyl group, halogen atom, nitro group, cyano group, C_{sub.1}-C_{sub.20} saturated hydrocarbylsulfinyl group, or C_{sub.1}-C_{sub.20} saturated hydrocarbylsulfonyl group. The saturated hydrocarbyl group, saturated hydrocarbyloxy group, saturated hydrocarbylcarbonyloxy group, saturated hydrocarbyloxyhydrocarbyl group, saturated hydrocarbylthiohydrocarbyl group, saturated hydrocarbylsulfinyl group, and saturated hydrocarbylsulfonyl group may be straight, branched or cyclic. When e₃ is 2 or more, a plurality of R^{sup.43} may be identical or different.

[0193] R^{sup.43} is preferably selected from halogen atoms such as chlorine, bromine and iodine; saturated hydrocarbyl groups such as methyl, ethyl, propyl, butyl, pentyl, hexyl, cyclopentyl, cyclohexyl, and structural isomers thereof; and saturated hydrocarbyloxy groups such as methoxy, ethoxy, propoxy, butoxy, pentyloxy, hexyloxy, cyclopentyloxy, cyclohexyloxy, and structural isomers of their hydrocarbon moiety. Inter alia, methoxy and ethoxy are most useful.

[0194] The saturated hydrocarbylcarbonyloxy group may be readily introduced into a polymer even after polymerization, by a chemical modification method and is advantageously utilized for fine adjustment of the solubility of the polymer in alkaline developer. Examples of the saturated hydrocarbylcarbonyloxy group include methylcarbonyloxy, ethylcarbonyloxy, propylcarbonyloxy, butylcarbonyloxy, penty carbonyloxy, hexylcarbonyloxy, cyclopentylcarbonyloxy, cyclohexylcarbonyloxy, benzoyloxy, and structural isomers of their hydrocarbon moiety. As long as the carbon count is equal to or less than 20, an appropriate effect of controlling or adjusting (typically reducing) the solubility of the polymer in alkaline developer is obtainable, and the generation of scum or development defects may be suppressed.

[0195] Of the foregoing preferred substituent groups, such substituent groups as chlorine, bromine, iodine, methyl, ethyl and methoxy are useful because the corresponding monomers may be readily prepared.

[0196] In formula (B6), A^{sup.3} is a single bond or C_{sub.1}-C_{sub.10} saturated hydrocarbylene group in which some constituent —CH_{sub.2}— may be replaced by —O—. The saturated hydrocarbylene group may be straight, branched or cyclic. Examples thereof include C_{sub.1}-C_{sub.10} alkanediyl groups such as methylene, ethane-1,2-diyl, propane-1,3-diyl, butane-1,4-diyl, pentane-1,5-diyl, hexane-1,6-diyl, and structural isomers thereof; C_{sub.3}-C_{sub.10} cyclic saturated hydrocarbylene groups such as cyclopropanediyl, cyclobutanediyl, cyclopentaned iyl, and cyclohexaned iyl; and combinations thereof. For the saturated hydrocarbylene group containing an ether bond, in case of e₁=1 in formula (B6), the ether bond may be incorporated at any position excluding the position between the α- and β-carbons relative to the ester oxygen. In case of e₁=0, the atom bonding to the backbone becomes an ether oxygen atom, and a second ether bond may be incorporated at any position excluding the position between the α- and β-carbons relative to the ether oxygen. Saturated hydrocarbylene groups having no more than 10 carbon atoms are desirable

because of a sufficient solubility in alkaline developer.

[0197] Preferred examples of the repeat units B6 wherein e1 is 0 and A.sup.3 is a single bond (meaning that the aromatic ring is directly bonded to the main chain of the polymer), that is, repeat units free of the linker: —C(=O)—O-A.sup.3- include units derived from styrene, 4-chlorostyrene, 4-bromostyrene, 4-methylstyrene, 4-methoxystyrene, 4-acetoxystyrene, 2-hydroxypropylstyrene, 2-vinylnaphthalene, and 3-vinylnaphthalene.

[0198] Preferred examples of the repeat units B6 wherein e1 is 1, that is, having the linker: —C(=O)—O-A.sup.3- are shown below, but not limited thereto. R.sup.A is as defined above.

##STR00621##

[0199] When repeat units of at least one type selected from repeat units B4 to B6 are incorporated, better performance is obtained because not only the aromatic ring possesses etch resistance, but the cyclic structure incorporated into the main chain also exerts the effect of improving resistance to etching and EB irradiation during pattern inspection step.

[0200] The repeat units B4 to B6 may be of one type or a combination of plural types.

[0201] The polymer B' may further comprise repeat units of at least one type selected from repeat units having the formula (B7), repeat units having the formula (B8), repeat units having the formula (B9), repeat units having the formula (B10), repeat units having the formula (B11), repeat units having the formula (B12), repeat units having the formula (B13), and repeat units having the formula (B14). Notably these repeat units are also referred to as repeat units B7 to B14. The polymer B' in this embodiment is more effective for restraining acid diffusion and ensures to form a pattern with improved resolution and reduced LER.

##STR00622## ##STR00623##

[0202] In formulae (B7) to (B14), R.sup.B is each independently hydrogen or methyl. Y.sup.1 is each independently a single bond, a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, naphthylene group, or C.sub.7-C.sub.18 group obtained by combining the foregoing, *-O—Y.sup.11—, *-C(=O)—O—Y.sup.11—, or *-C(=O)—NH—Y.sup.11—, wherein Y.sup.11 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, naphthylene group, or C.sub.7-C.sub.18 group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety. Y.sup.2 is each independently a single bond or **—Y.sup.21—C(=O)—O—, wherein Y.sup.21 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom. Y.sup.3 is each independently a single bond, methylene, ethylene, phenylene, fluorinated phenylene, trifluoromethyl-substituted phenylene group, *-O—Y.sup.31—, *-C(=O)—O—Y.sup.31—, or *-C(=O)—NH—Y.sup.31—, wherein Y.sup.31 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, fluorinated phenylene group, trifluoromethyl-substituted phenylene group, or C.sub.7-C.sub.20 group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety. Herein * designates a point of attachment to the carbon atom in the backbone and ** designates a point of attachment to the oxygen atom in the formula. Y.sup.4 is each independently a single bond or a C.sub.1-C.sub.30 hydrocarbylene group which may contain a heteroatom. The subscripts f1 and f2 are each independently 0 or 1. When Y.sup.4 is a single bond, f1 and f2 are 0.

[0203] In formulae (B7) and (B11), Xa.sup.- is a non-nucleophilic counter ion, examples of which include those described in JP-A 2010-113209 and JP-A 2007-145797.

[0204] In formulae (B8) and (B12) wherein Y.sup.2 is —Y.sup.21—C(=O)—O—, Y.sup.21 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom, examples of which are shown below, but not limited thereto.

##STR00624##

[0205] In formulae (B8) and (B12), R.sup.HF is hydrogen or trifluoromethyl. Examples of the repeat units B8 and B12 wherein R.sup.HF is hydrogen include those described in JP-A 2010-116550. Examples of the repeat units B8 and B12 wherein R.sup.HF is trifluoromethyl include those described in JP-A 2010-077404. Examples of the repeat units B8 and B12 include those

described in JP-A 2012-246265 and JP-A 2012-246426.

[0206] Preferred examples of the anion in the monomers from which repeat units B10 and B14 are derived are shown below, but not limited thereto.

##STR00625## ##STR00626## ##STR00627## ##STR00628## ##STR00629## ##STR00630##

[0207] In formulae (B7) to (B14), R^{sup.51} to R^{sup.68} are each independently halogen or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples of the halogen and hydrocarbyl group are as exemplified above for R^{sup.ct1} to R^{sup.ct5} in formulae (cation-1) and (cation-2). In the hydrocarbyl group, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some constituent —CH_{sub.2}— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy moiety, fluorine, chlorine, bromine, iodine, cyano moiety, nitro moiety, carbonyl moiety, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety.

[0208] A pair of R^{sup.51} and R^{sup.52} may bond together to form a ring with the sulfur atom to which they are attached. Also, R^{sup.53} and R^{sup.54}, R^{sup.56} and R^{sup.57}, or R^{sup.59} and R^{sup.60} may bond together to form a ring with the sulfur atom to which they are attached. Examples of the ring are as exemplified for the ring that R^{sup.ct1} and R^{sup.ct2} in formula (cation-1), taken together, form with the sulfur atom to which they are attached.

[0209] Examples of the sulfonium cation in repeat units B8 to B10 are as exemplified for the sulfonium cation having formula (cation-1). Examples of the iodonium cation in repeat units B12 to B14 are as exemplified for the iodonium cation having formula (cation-2).

[0210] The repeat units B7 to B14 are capable of generating an acid upon receipt of high-energy radiation. With the relevant units bound into a polymer, an appropriate control of acid diffusion becomes possible, and a pattern with reduced LER can be formed. Since the acid-generating unit is bound to a polymer, the phenomenon that acid volatilizes from the exposed region and re-deposits on the unexposed region during bake in vacuum is suppressed. This is effective for reducing LER and for suppressing unwanted negative-working reaction in the unexposed region for thereby reducing defects.

[0211] Each of the repeat units B7 to B14 may be of one type or a combination of plural types.

[0212] In the polymer B or B', (meth)acrylate and other repeat units having an adhesive group such as lactone structure or hydroxy group other than phenolic hydroxy may be incorporated for fine adjustment of properties of a resist film.

[0213] Examples of the (meth)acrylate unit having an adhesive group include repeat units having the following formulae (B15) to (B17), which are also referred to as repeat units B15 to B17. While these units do not exhibit acidity, they may be used as auxiliary units for providing adhesion to substrates or adjusting solubility.

##STR00631##

[0214] In formulae (B15) to (B17), R^{sup.A} is each independently hydrogen, fluorine, methyl or trifluoromethyl. R^{sup.71} is —O— or methylene. R^{sup.72} is hydrogen or hydroxy. R^{sup.73} is a C_{sub.1}-C_{sub.4} saturated hydrocarbyl group, and g is an integer of 0 to 3. Each of the repeat units B15 to B17 may be of one type or a combination of plural types.

[0215] In polymer B, an appropriate content of repeat units B1 is 30 to 95 mol %, more preferably 50 to 85 mol % for establishing a high contrast between a region which is exposed to high-energy radiation and turns negative and the unexposed region (which does not turn negative) for the purpose of achieving high resolution. An appropriate content of repeat units B2 and B3 is 5 to 70 mol %, more preferably 10 to 60 mol % for gaining the effect of promoting negative-working reaction. An appropriate content of repeat units B4 to B6 is 0 to 30 mol %, more preferably 3 to 20 mol % for gaining the effect of improving etch resistance. The other repeat units may be

incorporated in a range of 0 to 30 mol %, preferably 0 to 20 mol %.

[0216] Where the polymer B' is free of repeat units B7 to B14, the polymer B' preferably contains 25 to 95 mol %, more preferably 40 to 85 mol % of repeat units B1. An appropriate content of repeat units B4 to B6 is 0 to 30 mol %, more preferably 3 to 20 mol %. An appropriate content of repeat units B2 and B3 is 5 to 70 mol %, more preferably 10 to 60 mol %. The other repeat units may be incorporated in a range of 0 to 30 mol %, preferably 0 to 20 mol %.

[0217] Where the polymer B' contains repeat units B7 to B14, the polymer B' preferably contains 25 to 94.5 mol %, more preferably 36 to 85 mol % of repeat units B1. An appropriate content of repeat units B4 to B6 is 0 to 30 mol %, more preferably 3 to 20 mol %. An appropriate content of repeat units B2 and B3 is 5 to 70 mol %, more preferably 10 to 60 mol %. The total content of repeat units B1 to B6 is preferably 60 to 99.5 mol %. An appropriate content of repeat units B7 to B14 is 0.5 to 20 mol %, more preferably 1 to 10 mol %. The other repeat units may be incorporated in a range of 0 to 30 mol %, preferably 0 to 20 mol %.

[0218] It is noted that the repeat units B1 to B6 preferably account for at least 60 mol %, more preferably at least 70 mol %, even more preferably at least 80 mol % of the overall repeat units of the polymer. This range ensures that the polymer has necessary properties as the chemically amplified negative resist composition.

[0219] Preferably, the polymer B' comprises repeat units having the formula (B1-1), repeat units having the formula (B2-1), (B2-2) or (B3-1), and repeat units having the formula (B8).

##STR00632## ##STR00633##

[0220] Herein R.sup.A, R.sup.B, Y.sup.2, R.sup.32, R.sup.33, R.sup.35, R.sup.36, R.sup.53, R.sup.54, R.sup.55, R.sup.HF, a4, b4, and b6 are as defined above.

[0221] Where the polymer B' is used as the base polymer (B), it may be a mixture of a polymer free of repeat units B7 to B14 and a polymer comprising repeat units B7 to B14. In this embodiment, the polymer free of repeat units B7 to B14 is preferably used in an amount of 2 to 5,000 parts, more preferably 10 to 1,000 parts by weight per 100 parts by weight of the polymer comprising repeat units B7 to B14.

[0222] Reference is now made to the use of the chemically amplified negative resist composition in the fabrication of photomasks. The lithography of the advanced generation employs a coating film having a thickness of up to 150 nm, preferably up to 100 nm. Since an intense development process is often employed to minimize defects resulting from resist residues, the base polymer should preferably have a dissolution rate in alkaline developer (typically 2.38 wt % tetramethylammonium hydroxide (TMAH) aqueous solution) of up to 80 nm/sec, more preferably up to 50 nm/sec in order to form a small size pattern. When the chemically amplified negative resist composition is used in the EUV lithography process for fabricating an LSI chip from a wafer, for example, the coating film often has a thickness of up to 100 nm, in view of the necessity of patterning narrow lines of 50 nm or less. In consideration of the risk that the pattern of such thin film can be degraded by development, the polymer preferably has a dissolution rate of up to 80 nm/sec, more preferably up to 50 nm/sec.

[0223] The polymer may be synthesized by combining suitable monomers optionally protected with a protective group, copolymerizing them in the standard way, and effecting deprotection reaction if necessary. The copolymerization reaction is preferably radical or anionic polymerization though not limited thereto. For the polymerization reaction, reference may be made to WO 2006/121096, JP-A 2004-115630, JP-A 2008-102383, and JP-A 2008-304590.

[0224] The polymer should preferably have a Mw of 1,000 to 50,000, and more preferably 2,000 to 20,000. A Mw of at least 1,000 eliminates the risk that pattern features are rounded at their top, inviting degradations of resolution and LER. A Mw of up to 50,000 eliminates the risk that LER is increased when a pattern with a line width of up to 100 nm is formed. As used herein, Mw is measured by GPC versus polystyrene standards using tetrahydrofuran (THF) or dimethylformamide (DMF).

[0225] The polymer preferably has a narrow molecular weight distribution or dispersity (M_w/M_n) of 1.0 to 2.0, more preferably 1.0 to 1.8. A polymer with such a narrow dispersity eliminates the risk that foreign particles are left on the pattern after development and the pattern profile is aggravated.

(C) Crosslinker

[0226] When the base polymer (B) does not contain polymer B', the negative resist composition preferably comprises a crosslinker as component (C). When the base polymer (B) contains polymer B', a crosslinker need not be added.

[0227] Suitable crosslinkers which can be used herein include epoxy compounds, melamine compounds, guanamine compounds, glycoluril compounds and urea compounds having substituted thereon at least one group selected from among methylol, alkoxymethyl and acyloxymethyl groups, isocyanate compounds, azide compounds, and compounds having a double bond such as an alkenyloxy group. These compounds may be used as an additive or introduced into a polymer side chain as a pendant. Hydroxy-containing compounds may also be used as the crosslinker.

[0228] Of the foregoing crosslinkers, examples of suitable epoxy compounds include tris(2,3-epoxypropyl) isocyanurate, trimethylolmethane triglycidyl ether, trimethylolpropane triglycidyl ether, and triethylolethane triglycidyl ether.

[0229] Examples of the melamine compound include hexamethylol melamine, hexamethoxymethyl melamine, hexamethylol melamine compounds having 1 to 6 methylol groups methoxymethylated and mixtures thereof, hexamethoxyethyl melamine, hexaacyloxymethyl melamine, hexamethylol melamine compounds having 1 to 6 methylol groups acyloxymethylated and mixtures thereof.

[0230] Examples of the guanamine compound include tetramethylol guanamine, tetramethoxymethyl guanamine, tetramethylol guanamine compounds having 1 to 4 methylol groups methoxymethylated and mixtures thereof, tetramethoxyethyl guanamine, tetraacyloxyguanamine, tetramethylol guanamine compounds having 1 to 4 methylol groups acyloxymethylated and mixtures thereof.

[0231] Examples of the glycoluril compound include tetramethylol glycoluril, tetramethoxyglycoluril, tetramethoxymethyl glycoluril, tetramethylol glycoluril compounds having 1 to 4 methylol groups methoxymethylated and mixtures thereof, tetramethylol glycoluril compounds having 1 to 4 methylol groups acyloxymethylated and mixtures thereof.

[0232] Examples of the urea compound include tetramethylol urea, tetramethoxymethyl urea, tetramethylol urea compounds having 1 to 4 methylol groups methoxymethylated and mixtures thereof, and tetramethoxyethyl urea.

[0233] Suitable isocyanate compounds include tolylene diisocyanate, diphenylmethane diisocyanate, hexamethylene diisocyanate and cyclohexane diisocyanate.

[0234] Suitable azide compounds include 1,1'-biphenyl-4,4'-bisazide, 4,4'-methylidenebisazide, and 4,4'-oxybisazide.

[0235] Examples of the alkenyloxy-containing compound include ethylene glycol divinyl ether, triethylene glycol divinyl ether, 1,2-propanediol divinyl ether, 1,4-butanediol divinyl ether, tetramethylene glycol divinyl ether, neopentyl glycol divinyl ether, trimethylol propane trivinyl ether, hexanediol divinyl ether, 1,4-cyclohexanediol divinyl ether, pentaerythritol trivinyl ether, pentaerythritol tetravinyl ether, sorbitol tetravinyl ether, sorbitol pentavinyl ether, and trimethylol propane trivinyl ether.

[0236] An appropriate amount of the crosslinker (C) used is 0.1 to 50 parts, and more preferably 1 to 30 parts by weight per 80 parts by weight of the base polymer (B). As long as the amount of the crosslinker is in the range, the risk of resolution being reduced by forming bridges between pattern features is mitigated. The crosslinkers may be used alone or in admixture.

(D) Fluorinated Polymer

[0237] The negative resist composition may further comprise a fluorinated polymer for the purposes of enhancing contrast, preventing chemical flare of acid upon exposure to high-energy

radiation, preventing mixing of acid from an anti-charging film in the step of coating an anti-charging film-forming material on a resist film, and suppressing unexpected unnecessary pattern degradation. The fluorinated polymer contains repeat units of at least one type selected from repeat units having the formula (D1), repeat units having the formula (D2), repeat units having the formula (D3), and repeat units having the formula (D4), and may contain repeat units of at least one type selected from repeat units having the formula (D5) and repeat units having the formula (D6). It is noted that repeat units having formulae (D1), (D2), (D3), (D4), (D5), and (D6) are also referred to as repeat units D1, D2, D3, D4, D5, and D6, respectively, hereinafter. Since the fluorinated polymer also has a surface active function, it can prevent insoluble residues from re-depositing onto the substrate during the development step and is thus effective for preventing development defects.

##STR00634##

[0238] In formulae (D1) to (D6), x is an integer of 1 to 3, y is an integer satisfying: $0 \leq y \leq 5 + 2z - x$, z is 0 or 1, and g is an integer of 1 to 3. R.sup.C is each independently hydrogen, fluorine, methyl or trifluoromethyl. R.sup.D is each independently hydrogen or methyl. R.sup.101, R.sup.102, R.sup.104 and R.sup.105 are each independently hydrogen or a C.sub.1-C.sub.10 saturated hydrocarbyl group. R.sup.103, R.sup.106, R.sup.107 and R.sup.108 are each independently hydrogen, a C.sub.1-C.sub.15 hydrocarbyl group or fluorinated hydrocarbyl group, or an acid labile group, with the proviso that an ether bond or carbonyl moiety may intervene in a carbon-carbon bond in the hydrocarbyl groups or fluorinated hydrocarbyl groups represented by R.sup.103, R.sup.106, R.sup.107 and R.sup.108. R.sup.109 is hydrogen or a C.sub.1-C.sub.5 straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond. R.sup.110 is a C.sub.1-C.sub.5 straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond. R.sup.111 is a C.sub.1-C.sub.20 saturated hydrocarbyl group in which at least one hydrogen is substituted by fluorine and some constituent —CH.sub.2— may be replaced by an ester bond or ether bond. Z.sup.1 is a C.sub.1-C.sub.20 (g+1)-valent hydrocarbon group or C.sub.1-C.sub.20 (g+1)-valent fluorinated hydrocarbon group. Z.sup.2 is a single bond, *—C(=O)—O— or *—C(=O)—NH— wherein * designates a point of attachment to the carbon atom in the backbone. Z.sup.3 is a single bond, —O—, *—C(=O)—O—Z.sup.31—Z.sup.32— or *—C(=O)—NH—Z.sup.31—Z.sup.32—, wherein Z.sup.31 is a single bond or a C.sub.1-C.sub.10 saturated hydrocarbylene group, Z.sup.32 is a single bond, ester bond, ether bond or sulfonamide bond, and * designates a point of attachment to the carbon atom in the backbone.

[0239] In formulae (D1) and (D2), examples of the C.sub.1-C.sub.10 saturated hydrocarbyl group represented by R.sup.101, R.sup.102, R.sup.104 and R.sup.105 include C.sub.1-C.sub.10 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, n-hexyl, n-heptyl, n-octyl, n-nonyl, and n-decyl, and C.sub.3-C.sub.10 cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, adamantyl, and norbornyl. Inter alia, C.sub.1-C.sub.6 saturated hydrocarbyl groups are preferred.

[0240] In formulae (D1) to (D4), the C.sub.1-C.sub.15 hydrocarbyl group represented by R.sup.103, R.sup.106, R.sup.107 and R.sup.108 may be straight, branched or cyclic and examples thereof include C.sub.1-C.sub.15 alkyl, C.sub.2-C.sub.15 alkenyl and C.sub.2-C.sub.15 alkynyl groups, with the alkyl groups being preferred. Suitable alkyl groups include methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, n-pentyl, n-hexyl, n-heptyl, n-octyl, n-nonyl, n-decyl, n-undecyl, n-dodecyl, n-tridecyl, n-tetradecyl and n-pentadecyl. The fluorinated hydrocarbyl groups correspond to the foregoing hydrocarbyl groups in which some or all carbon-bonded hydrogen atoms are substituted by fluorine atoms.

[0241] In formula (D4), examples of the C.sub.1-C.sub.20 (g+1)-valent hydrocarbon group Z.sup.1 include the foregoing C.sub.1-C.sub.20 alkyl groups and C.sub.3-C.sub.20 cyclic saturated hydrocarbyl groups, with g number of hydrogen atoms being eliminated. Examples of the C.sub.1-

C.sub.20 (g+1)-valent fluorinated hydrocarbon group Z.sup.1 include the foregoing (g+1)-valent hydrocarbon groups in which at least one hydrogen atom is substituted by fluorine.

[0242] Examples of the repeat units D1 to D4 are given below, but not limited thereto. Herein R.sup.C is as defined above.

##STR00635## ##STR00636## ##STR00637## ##STR00638##

[0243] In formula (D5), examples of the C.sub.1-C.sub.5 hydrocarbyl groups R.sup.109 and R.sup.110 include alkyl, alkenyl and alkynyl groups, with the alkyl groups being preferred. Suitable alkyl groups include methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, and n-pentyl. In these groups, a moiety containing a heteroatom such as oxygen, sulfur or nitrogen may intervene in a carbon-carbon bond.

[0244] In formula (D5), —OR.sup.109 is preferably a hydrophilic group. In this case, R.sup.109 is preferably hydrogen or a C.sub.1-C.sub.5 alkyl group in which oxygen intervenes in a carbon-carbon bond.

[0245] In formula (D5), Z.sup.2 is preferably *—C(=O)—O— or *—C(=O)—NH—. Also preferably R.sup.D is methyl. The inclusion of carbonyl in Z.sup.2 enhances the ability to trap the acid originating from the anti-charging film. A polymer wherein R.sup.D is methyl is a robust polymer having a high glass transition temperature which is effective for suppressing acid diffusion. As a result, the resist film is improved in stability with time, and neither resolution nor pattern profile is degraded.

[0246] Examples of the repeat unit D5 are given below, but not limited thereto. Herein R.sup.D is as defined above.

##STR00639## ##STR00640## ##STR00641##

[0247] In formula (D6), the C.sub.1-C.sub.10 saturated hydrocarbylene group Z.sup.3 may be straight, branched or cyclic and examples thereof include methanediyl, ethane-1,1-diyl, ethane-1,2-diyl, propane-1,1-diyl, propane-1,2-diyl, propane-1,3-diyl, propane-2,2-diyl, butane-1,1-diyl, butane-1,2-diyl, butane-1,3-diyl, butane-2,3-diyl, butane-1,4-diyl, and 1,1-dimethylethane-1,2-diyl.

[0248] The C.sub.1-C.sub.20 saturated hydrocarbyl group having at least one hydrogen substituted by fluorine, represented by R.sup.111, may be straight, branched or cyclic and examples thereof include C.sub.1-C.sub.20 alkyl groups and C.sub.3-C.sub.20 cyclic saturated hydrocarbyl groups in which at least one hydrogen is substituted by fluorine.

[0249] Examples of the repeat unit D6 are given below, but not limited thereto. Herein R.sup.D is as defined above.

##STR00642## ##STR00643## ##STR00644## ##STR00645## ##STR00646## ##STR00647##

[0250] The repeat units D1 to D4 are preferably incorporated in an amount of 15 to 95 mol %, more preferably 20 to 85 mol % based on the overall repeat units of the fluorinated polymer. The repeat unit D5 and/or D6 is preferably incorporated in an amount of 5 to 85 mol %, more preferably 15 to 80 mol % based on the overall repeat units of the fluorinated polymer. Each of repeat units D1 to D6 may be used alone or in admixture.

[0251] The fluorinated polymer may comprise additional repeat units as well as the repeat units D1 to D6. Suitable additional repeat units include those described in U.S. Pat. No. 9,091,918 (JP-A 2014-177407, paragraphs [0046]-[0078]). When the fluorinated polymer comprises additional repeat units, their content is preferably up to 50 mol % based on the overall repeat units.

[0252] The fluorinated polymer may be synthesized by combining suitable monomers optionally protected with a protective group, copolymerizing them in the standard way, and effecting deprotection reaction if necessary. The copolymerization reaction is preferably radical or anionic polymerization though not limited thereto. For the polymerization reaction, reference may be made to JP-A 2004-115630.

[0253] The fluorinated polymer should preferably have a Mw of 2,000 to 50,000, and more preferably 3,000 to 20,000. A fluorinated polymer with a Mw of less than 2,000 helps acid diffusion, degrading resolution and detracting from age stability. A polymer with too high Mw has

a reduced solubility in solvent, with a risk of leaving coating defects. The fluorinated polymer preferably has a dispersity (Mw/Mn) of 1.0 to 2.2, more preferably 1.0 to 1.7.

[0254] In the negative resist composition, the fluorinated polymer (D) is preferably used in an amount of 0.01 to 30 parts, more preferably 0.1 to 20 parts, even more preferably 0.5 to 10 parts by weight per 80 parts by weight of the base polymer (B). The fluorinated polymer may be used alone or in admixture.

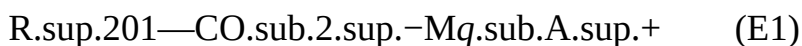
(E) Quencher

[0255] The negative resist composition preferably contains a quencher as component (E). As used herein, the quencher refers to a compound capable of trapping the acid generated by the PAG in the resist composition to prevent the acid from diffusing to the unexposed region for thereby forming the desired pattern.

[0256] The quencher is typically selected from conventional basic compounds. Conventional basic compounds include primary, secondary, and tertiary aliphatic amines, mixed amines, aromatic amines, heterocyclic amines, nitrogen-containing compounds with carboxy group, nitrogen-containing compounds with sulfonyl group, nitrogen-containing compounds with hydroxy group, nitrogen-containing compounds with hydroxyphenyl group, alcoholic nitrogen-containing compounds, amide derivatives, imide derivatives, and carbamate derivatives. Also included are primary, secondary, and tertiary amine compounds, specifically amine compounds having a hydroxy group, ether bond, ester bond, lactone ring, cyano group, or sulfonate ester bond as described in JP-A 2008-111103, paragraphs [0146]-[0164], and compounds having a carbamate group as described in JP 3790649. Inter alia, tris[2-(methoxymethoxy)ethyl]amine, tris[2-(methoxymethoxy)ethyl]amine-N-oxide, dibutylaminobenzoic acid, morpholine derivatives, and imidazole derivatives are preferred. Addition of a basic compound is effective for further suppressing the diffusion rate of acid in the resist film or correcting the pattern profile.

[0257] Onium salts such as sulfonium, iodonium and ammonium salts of carboxylic acids which are not fluorinated at α -position as described in U.S. Pat. No. 8,795,942 (JP-A 2008-158339) may also be used as the quencher. While an α -fluorinated sulfonic acid, imide acid, and methide acid are necessary to deprotect the acid labile group, an α -non-fluorinated carboxylic acid is released by salt exchange with an α -non-fluorinated onium salt. An α -non-fluorinated carboxylic acid functions as a quencher because it does not induce substantial deprotection reaction.

[0258] Examples of the onium salt of α -non-fluorinated carboxylic acid include compounds having the formula (E1).



[0259] In formula (E1), R.sub.201 is hydrogen or a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom, exclusive of the hydrocarbyl group in which the hydrogen bonded to the carbon atom at α -position of the carboxy group is substituted by fluorine or fluoroalkyl.

[0260] The hydrocarbyl group R.sub.201 may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.40 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, sec-butyl, tert-butyl, n-pentyl, tert-pentyl, n-hexyl, n-octyl, 2-ethylhexyl, n-nonyl, n-decyl; C.sub.3-C.sub.40 cyclic saturated hydrocarbyl groups such as cyclopentyl, cyclohexyl, cyclopentylmethyl, cyclopentylethyl, cyclopentylbutyl, cyclohexylmethyl, cyclohexylethyl, cyclohexylbutyl, norbornyl, tricyclo[5.2.1.0^{sup}.2,6]decyl, adamantyl, and adamantylmethyl; C.sub.2-C.sub.40 alkenyl groups such as vinyl, allyl, propenyl, butenyl and hexenyl; C.sub.3-C.sub.40 cyclic unsaturated aliphatic hydrocarbyl groups such as cyclohexenyl; C.sub.6-C.sub.40 aryl groups such as phenyl, naphthyl, alkylphenyl groups (e.g., 2-methylphenyl, 3-methylphenyl, 4-methylphenyl, 4-ethylphenyl, 4-tert-butylphenyl, 4-n-butylphenyl), di- or trialkylphenyl groups (e.g., 2,4-dimethylphenyl and 2,4,6-triisopropylphenyl), alkyl-naphthyl groups (e.g., methyl-naphthyl and ethyl-naphthyl), dialkyl-naphthyl groups (e.g., dimethyl-naphthyl and diethyl-naphthyl); and C.sub.7-C.sub.40 aralkyl groups such as benzyl, 1-phenylethyl and 2-

phenylethyl.

[0261] In the foregoing hydrocarbonyl groups, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some constituent —CH₂— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy moiety, cyano moiety, carbonyl moiety, ether bond, thioether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety. Suitable heteroatom-containing hydrocarbonyl groups include heteroaryl groups such as thienyl; alkoxyphenyl groups such as 4-hydroxyphenyl, 4-methoxyphenyl, 3-methoxyphenyl, 2-methoxyphenyl, 4-ethoxyphenyl, 4-tert-butoxyphenyl, 3-tert-butoxyphenyl; alkoxynaphthyl groups such as methoxynaphthyl, ethoxynaphthyl, n-propoxynaphthyl and n-butoxynaphthyl; dialkoxynaphthyl groups such as dimethoxynaphthyl and diethoxynaphthyl; and aryloxyalkyl groups, typically 2-aryl-2-oxoethyl groups such as 2-phenyl-2-oxoethyl, 2-(1-naphthyl)-2-oxoethyl and 2-(2-naphthyl)-2-oxoethyl.

[0262] In formula (E1), Mq_A⁺ is an onium cation. The onium cation is preferably selected from sulfonium, iodonium and ammonium cations, more preferably sulfonium and iodonium cations. Exemplary sulfonium cations are as exemplified above for the sulfonium cation having formula (cation-1). Exemplary iodonium cations are as exemplified above for the iodonium cation having formula (cation-2).

[0263] Examples of the anion in the onium salt having formula (E1) are shown below, but not limited thereto.

##STR00648## ##STR00649## ##STR00650## ##STR00651##

[0264] A sulfonium salt of iodized benzene ring-containing carboxylic acid having the formula (E2) is also useful as the quencher.

##STR00652##

[0265] In formula (E2), s is an integer of 1 to 5, t is an integer of 0 to 3, s+u is from 1 to 5, and u is an integer of 1 to 3.

[0266] In formula (E2), R²¹¹ is hydroxy, fluorine, chlorine, bromine, amino, nitro, cyano, or a C₁-C₆ saturated hydrocarbonyl, C₁-C₆ saturated hydrocarbonyloxy, C₂-C₆ saturated hydrocarbonylcarbonyloxy or C₁-C₄ saturated hydrocarbonylsulfonyloxy group, in which some or all hydrogen may be substituted by halogen, or —N(R^{211A})—C(=O)—R^{211B}, or —N(R^{211A})—C(=O)—O—R^{211B}. R^{211A} is hydrogen or a C₁-C₆ saturated hydrocarbonyl group. R^{211B} is a C₁-C₆ saturated hydrocarbonyl or C₂-C₈ unsaturated aliphatic hydrocarbonyl group. When t and/or u is 2 or 3, a plurality of R²¹¹ may be the same or different.

[0267] In formula (E2), L¹ is a single bond, or a C₁-C₂₀ (u+1)-valent linking group which may contain at least one moiety selected from ether bond, carbonyl moiety, ester bond, amide bond, sultone ring, lactam ring, carbonate bond, halogen, hydroxy moiety, and carboxy moiety. The saturated hydrocarbonyl, saturated hydrocarbonyloxy, saturated hydrocarbonylcarbonyloxy, and saturated hydrocarbonylsulfonyloxy groups may be straight, branched or cyclic.

[0268] In formula (E2), R²¹², R²¹³ and R²¹⁴ are each independently halogen, or a C₁-C₂₀ hydrocarbonyl group which may contain a heteroatom. The hydrocarbonyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C₁-C₂₀ alkyl, C₂-C₂₀ alkenyl, C₆-C₂₀ aryl, and C₇-C₂₀ aralkyl groups. In these groups, some or all hydrogen may be substituted by hydroxy moiety, carboxy moiety, halogen, oxo moiety, cyano moiety, nitro moiety, sultone ring, sulfo moiety, or sulfonium salt-containing moiety, or some constituent —CH₂— may be replaced by an ether bond, ester bond, carbonyl moiety, amide bond, carbonate bond or sulfonate ester bond. Also R²¹² and R²¹³ may bond together to form a ring with the sulfur atom to which they are attached.

[0269] Examples of the compound having formula (E2) include those described in U.S. Pat. No. 10,295,904 (JP-A 2017-219836). These compounds exert a sensitizing effect due to remarkable

absorption and an acid diffusion-controlling effect.

[0270] A nitrogen-containing carboxylic acid salt compound having the formula (E3) is also useful as the quencher.

##STR00653##

[0271] In formula (E3), R^{sup.221} to R^{sup.224} are each independently hydrogen, -L^{sup.2}-CO₂^{sup.}, or a C₁₋₂₀ hydrocarbyl group which may contain a heteroatom. R^{sup.221} and R^{sup.222}, R^{sup.222} and R^{sup.223}, or R^{sup.223} and R^{sup.224} may bond together to form a ring with the carbon atom to which they are attached. L^{sup.2} is a single bond or a C₁₋₂₀ hydrocarbylene group which may contain a heteroatom. R^{sup.225} is hydrogen or a C₁₋₂₀ hydrocarbyl group which may contain a heteroatom.

[0272] In formula (E3), the ring R' is a C₂₋₆ ring containing the carbon and nitrogen atoms in the formula, in which some or all of the carbon-bonded hydrogen atoms may be substituted by a C₁₋₂₀ hydrocarbyl group or -L^{sup.2}-CO₂^{sup.} and in which some-CH₂— may be replaced by sulfur, oxygen or nitrogen. The ring may be alicyclic or aromatic and is preferably a 5- or 6-membered ring. Suitable rings include pyridine, pyrrole, pyrrolidine, piperidine, pyrazole, imidazoline, pyridazine, pyrimidine, pyrazine, imidazole, oxazole, thiazole, morpholine, thiazine, and triazole rings.

[0273] The carboxylic onium salt having formula (E3) has at least one -L^{sup.2}-CO₂^{sup.}. That is, at least one of R^{sup.221} to R^{sup.224} is -L^{sup.2}-CO₂^{sup.}, and/or at least one of hydrogen atoms bonded to carbon atoms in the ring R^{sup.r} is substituted by -L^{sup.2}-CO₂^{sup.}.

[0274] In formula (E3), M_q^{sub.B}^{sup.+} is a sulfonium, iodonium or ammonium cation, with the sulfonium cation being preferred. Examples of the sulfonium cation are as exemplified above for the sulfonium cation having formula (cation-1).

[0275] Examples of the anion in the compound having formula (E3) are shown below, but not limited thereto.

##STR00654## ##STR00655## ##STR00656## ##STR00657## ##STR00658## ##STR00659##

[0276] Weak acid betaine compounds are also useful as the quencher. Non-limiting examples thereof are shown below.

##STR00660## ##STR00661##

[0277] Also useful are quenchers of polymer type as described in U.S. Pat. No. 7,598,016 (JP-A 2008-239918). The polymeric quencher segregates at the resist surface after coating and thus enhances the rectangularity of resist pattern. When a protective film is applied as is often the case in the immersion lithography, the polymeric quencher is also effective for preventing a film thickness loss of resist pattern or rounding of pattern top.

[0278] When used, the quencher (E) is preferably added in an amount of 0 to 50 parts, more preferably 0.1 to 40 parts by weight per 80 parts by weight of the base polymer (B). The quencher may be used alone or in admixture.

[0279] In the embodiment wherein the chemically amplified negative resist composition contains the acid generator (A) and the quencher (E), the acid generator (A) and the quencher (E) are preferably present in a weight ratio (A/E) of less than 6/1, more preferably less than 5/1, even more preferably less than 4/1. As long as the ratio of acid generator (A) to quencher (E) is in the range, it is possible to fully suppress acid diffusion, leading to improved resolution and dimensional uniformity.

(F) Organic Solvent

[0280] The chemically amplified negative resist composition may further comprise an organic solvent as component (F). The organic solvent used herein is not particularly limited as long as the components are soluble therein. Examples of the organic solvent are described in JP-A 2008-111103, paragraphs to (U.S. Pat. No. 7,537,880). Specifically, exemplary solvents include ketones such as cyclohexanone, cyclopentanone, methyl-2-n-pentyl ketone and 2-heptanone; alcohols such as 3-methoxybutanol, 3-methyl-3-methoxybutanol, 1-methoxy-2-propanol, 1-ethoxy-2-propanol,

and diacetone alcohol; ethers such as propylene glycol monomethyl ether (PGME), ethylene glycol monomethyl ether, propylene glycol monoethyl ether, ethylene glycol monoethyl ether, propylene glycol dimethyl ether, and diethylene glycol dimethyl ether; esters such as propylene glycol monomethyl ether acetate (PGMEA), propylene glycol monoethyl ether acetate, ethyl lactate (EL), ethyl pyruvate, butyl acetate, methyl 3-methoxypropionate, ethyl 3-ethoxypropionate, t-butyl acetate, t-butyl propionate, and propylene glycol mono-t-butyl ether acetate; and lactones such as γ -butyrolactone (GBL), and mixtures thereof.

[0281] Of the above organic solvents, it is recommended to use 1-ethoxy-2-propanol, PGMEA, PGME, cyclohexanone, EL, GBL, and mixtures thereof.

[0282] In the negative resist composition, the organic solvent (F) is preferably used in an amount of 200 to 10,000 parts, more preferably 400 to 6,000 parts by weight per 80 parts by weight of the base polymer (B). The organic solvent may be used alone or in admixture.

(G) Surfactant

[0283] The negative resist composition may contain any conventional surfactants for facilitating to coat the composition to the substrate. Exemplary surfactants include PF-636 (Omnova Solutions Inc.) and FC-4430 (3M) as well as a number of known surfactants as described in JP-A 2004-115630. Any suitable one may be chosen therefrom. The amount of the surfactant (G) added is preferably 0 to 5 parts by weight per 80 parts by weight of the base polymer (B). The surfactant may be used alone or in admixture.

Process

[0284] Another embodiment of the invention is a resist pattern forming process comprising the steps of applying the negative resist composition defined above onto a substrate to form a resist film thereon, exposing the resist film patternwise to high-energy radiation, and developing the resist film in an alkaline developer to form a resist pattern.

[0285] Pattern formation using the negative resist composition of the invention may be performed by well-known lithography processes. In general, the resist composition is first applied onto a substrate for IC fabrication (e.g., Si, SiO, SiO.sub.2, SiN, SiON, TiN, WSi, BPSG, SOG, organic antireflective coating, etc.) or a substrate for mask circuit fabrication (e.g., Cr, CrO, CrON, MoSi.sub.2, Si, SiO, SiO.sub.2, SiON, SiONC, CoTa, NiTa, TaBN, SnO.sub.2, etc.) by a suitable coating technique such as spin coating. The coating is prebaked on a hotplate preferably at a temperature of 60 to 150° C. for 1 to 20 minutes, more preferably at 80 to 140° C. for 1 to 10 minutes to form a resist film of 0.03 to 2 μ m thick.

[0286] Then the resist film is exposed patternwise to high-energy radiation such as UV, deep-UV, excimer laser (KrF, ArF), EUV, x-ray, γ -ray or synchrotron radiation or EB. The resist composition of the invention is especially effective in the EUV or EB lithography.

[0287] On use of UV, deep-UV, EUV, excimer laser, x-ray, γ -ray or synchrotron radiation as the high-energy radiation, the resist film is exposed through a mask having a desired pattern, preferably in a dose of 1 to 500 mJ/cm.sup.2, more preferably 10 to 400 mJ/cm.sup.2. On use of EB, a pattern may be written directly in a dose of preferably 1 to 500 μ C/cm.sup.2, more preferably 10 to 400 μ C/cm.sup.2.

[0288] The exposure may be performed by conventional lithography whereas the immersion lithography of holding a liquid, typically water between the mask and the resist film may be employed if desired. In the case of immersion lithography, a protective film which is insoluble in water may be used.

[0289] The resist film is then baked (PEB) on a hotplate preferably at 60 to 150° C. for 1 to 20 minutes, more preferably at 80 to 140° C. for 1 to 10 minutes.

[0290] Thereafter, the resist film is developed with a developer in the form of an aqueous base solution, for example, 0.1 to 5 wt %, preferably 2 to 3 wt % aqueous solution of tetramethylammonium hydroxide (TMAH) preferably for 0.1 to 3 minutes, more preferably 0.5 to 2 minutes by conventional techniques such as dip, puddle and spray techniques. In this way, a desired

resist pattern is formed on the substrate.

[0291] From the negative resist composition, a pattern with a high resolution and minimal LER can be formed. The resist composition is effectively applicable to a substrate, specifically a substrate having a surface layer of material to which a resist film is less adherent and which is likely to invite pattern stripping or pattern collapse, and particularly a substrate having sputter deposited on its outermost surface metallic chromium or a chromium compound containing at least one light element selected from oxygen, nitrogen and carbon or a substrate having an outermost surface layer of SiO, SiO.sub.x, or a tantalum, molybdenum, cobalt, nickel, tungsten or tin compound. The substrate to which the negative resist composition is applied is most typically a photomask blank which may be either of transmission or reflection type.

[0292] The mask blank of transmission type is typically a photomask blank having a light-shielding film of chromium-based material. It may be either a photomask blank for binary masks or a photomask blank for phase shift masks. In the case of the binary mask-forming photomask blank, the light-shielding film may include an antireflection layer of chromium-based material and a light-shielding layer. In one example, the antireflection layer on the surface layer side is entirely composed of a chromium-based material. In an alternative example, only a surface side portion of the antireflection layer on the surface layer side is composed of a chromium-based material and the remaining portion is composed of a silicon compound-based material which may contain a transition metal. In the case of the phase shift mask-forming photomask blank, it may include a phase shift film and a chromium-based light-shielding film thereon.

[0293] Photomask blanks having an outermost layer of chromium base material are well known as described in JP-A 2008-026500 and JP-A 2007-302873 and the references cited therein. Although the detail description is omitted herein, the following layer construction may be employed when a light-shielding film including an antireflective layer and a light-shielding layer is composed of chromium base materials.

[0294] In the example where a light-shielding film including an antireflective layer and a light-shielding layer is composed of chromium base materials, layers may be stacked in the order of an antireflective layer and a light-shielding layer from the outer surface side, or layers may be stacked in the order of an antireflective layer, a light-shielding layer, and an antireflective layer from the outer surface side. Each of the antireflective layer and the light-shielding layer may be composed of multiple sub-layers. When the sub-layers have different compositions, the composition may be graded discontinuously or continuously from sub-layer to sub-layer. The chromium base material used herein may be metallic chromium or a material consisting of metallic chromium and a light element such as oxygen, nitrogen or carbon. Examples used herein include metallic chromium, chromium oxide, chromium nitride, chromium carbide, chromium oxynitride, chromium oxycarbide, chromium nitride carbide, and chromium oxide nitride carbide.

[0295] The mask blank of reflection type includes a substrate, a multilayer reflective film formed on one major surface (front surface) of the substrate, for example, a multilayer reflective film of reflecting exposure radiation such as EUV radiation, and an absorber film formed on the multilayer reflective film, for example, an absorber film of absorbing exposure radiation such as EUV radiation to reduce reflectivity. From the reflection type mask blank (reflection type mask blank for EUV lithography), a reflection type mask (reflection type mask for EUV lithography) having an absorber pattern (patterned absorber film) formed by patterning the absorber film is produced. The EUV radiation used in the EUV lithography has a wavelength of 13 to 14 nm, typically about 13.5 nm.

[0296] The multilayer reflective film is preferably formed contiguous to one major surface of a substrate. An underlay film may be disposed between the substrate and the multilayer reflective film as long as the benefits of the invention are not lost. The absorber film may be formed contiguous to the multilayer reflective film while a protective film (protective film for the multilayer reflective film) may be disposed between the multilayer reflective film and the absorber

film, preferably contiguous to the multilayer reflective film, more preferably contiguous to the multilayer reflective film and the absorber film. The protective film is used for protecting the multilayer reflective film in a cleaning, tailoring or otherwise processing step. Also preferably, the protective film has an additional function of protecting the multilayer reflective film or preventing the multilayer reflective film from oxidation during the step of patterning the absorber film by etching. Besides, an electroconductive film, which is used for electrostatic chucking of the reflection type mask to an exposure tool, may be disposed below the other major surface (back side surface) which is opposed to the one major surface of the substrate, preferably contiguous to the other major surface. It is provided herein that a substrate has one major surface which is a front or upper side surface and another major surface which is a back or lower side surface. The terms “front and back” sides or “upper and lower” sides are used for the sake of convenience. One or another major surface may be either of the two major surfaces (film-bearing surfaces) of a substrate, and in this sense, front and back or upper and lower are exchangeable. Specifically, the multilayer reflective film may be formed by any of the methods of JP-A 2021-139970 and the references cited therein.

[0297] The resist pattern forming process is successful in forming patterns having a very high resolution, reduced LER, rectangularity, and fidelity even on a substrate (typically mask blank of transmission or reflection type) whose outermost surface is made of a material tending to affect resist pattern profile such as a chromium, silicon or tantalum-containing material.

EXAMPLES

[0298] Examples of the invention are given below by way of illustration and not by way of limitation. The abbreviation “pbw” is parts by weight. For copolymers, the compositional ratio is a molar ratio and Mw is determined by GPC versus polystyrene standards using THF or DMF solvent. Analysis is made by IR and proton-NMR spectroscopy and time-of-flight mass spectrometry (TOF-MS). The analyzers used are shown below.

[0299] IR: NICOLET 6700 by Thermo Fisher Scientific Inc.

[0300] ¹H-NMR: ECA-500 by JEOL Ltd.

[0301] MALDI TOF-MS: S3000 by JEOL Ltd.

[1] Synthesis of Onium Salts

[Synthesis Example 1-1] Synthesis of PAG-1

##STR00662##

(1) Synthesis of Intermediate In-1

[0302] In nitrogen atmosphere, a reactor was charged with 6.3 g of reactant SM-1, 9.8 g of reactant SM-2, 0.3 g of DMAP, and 50 g of methylene chloride and cooled in an ice bath. While the internal temperature of the reactor was kept below 20° C., 5.8 g of 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride in powder form was added. Thereafter, the reactor was allowed to reach room temperature, at which the reaction solution was aged for 12 hours. At the end of aging, water was added to quench the reaction, followed by ordinary aqueous work-up. The subsequent steps of distilling off the solvent and washing the residue with diisopropyl ether gave 13.9 g of Intermediate In-1 as oily matter (yield 92%).

(2) Synthesis of Onium Salt PAG-1

[0303] In nitrogen atmosphere, a reactor was charged with 13.9 g of Intermediate In-1, 7.8 g of reactant SM-3, 50 g of methylene chloride, and 30 g of water, which were stirred for 15 minutes. The organic layer was taken out, washed with water, and concentrated under reduced pressure. To the concentrate, 50 g of methyl isobutyl ketone was added, followed by azeotropic dewatering. Diisopropyl ether was then added for recrystallization, obtaining 15.5 g of the target onium salt, PAG-1 as white crystals (yield 92%).

[0304] PAG-1 was analyzed by IR spectroscopy and TOF-MS, with the data shown below. FIG. 1 is the ¹H-NMR/DMSO-d₆ spectrum of PAG-1.

[0305] IR (D-ATR): ν=3447, 3064, 3022, 2971, 1751, 1626, 1594, 1583, 1507, 1464, 1442, 1379,

1368, 1322, 1272, 1204, 1159, 1132, 1113, 1064, 1040, 1021, 998, 905, 886, 832, 816, 762, 706, 684, 632, 601, 577, 534, 493 cm.sup.-1

[0306] MALDI TOF-MS: [0307] positive M.sup.+ 277 (corresponding to C.sub.18H.sub.13OS.sup.+) [0308] negative M.sup.- 455 (corresponding to C.sub.27H.sub.19O.sub.5S.sup.-)

[Synthesis Examples 1-2 to 1-8] Synthesis of Onium Salts PAG-2 to PAG-8

[0309] Onium salts PAG-2 to PAG-8 shown below were synthesized using corresponding reactants and well-known organic synthesis reactions.

##STR00663## ##STR00664##

[2] Preparation of Chemically Amplified Negative Resist Compositions

Examples 1-1 to 1-54 and Comparative Examples 1-1 to 1-10

[0310] Chemically amplified negative resist compositions (R-1 to R-54, CR-1 to CR-10) were prepared by dissolving selected components in an organic solvent in accordance with the formulation shown in Tables 1 to 3, and filtering the solution through a UPE filter and/or nylon filter with a pore size of 10 nm, 5 nm, 3 nm or 1 nm. The organic solvent was a mixture of 790 pbw of PGMEA, 1,580 pbw of EL, and 1,580 pbw of PGME. To some compositions, fluorinated polymer (Polymers FP-1 to FP-5) as additive, tetramethoxymethylglycoluril (TMGU) as crosslinker, and PF-636 (Omnova Solutions Inc.) as surfactant were added.

TABLE-US-00001 TABLE 1 Photoacid Fluorinated Resist generator Polymer Polymer Quencher Crosslinker polymer Surfactant composition (pbw) 1 (pbw) 2 (pbw) (pbw) (pbw) (pbw) (pbw)
Example 1-1 R-1 PAG-1 P-1 — Q-1 TMGU — PF-636 (10) (80) (4.0) (8.154) (0.075) 1-2 R-2 PAG-1 P-1 — Q-1 TMGU FP-1 PF-636 (10) (80) (5.0) (8.154) (3) (0.075) 1-3 R-3 PAG-1 P-2 — Q-2 TMGU — PF-636 (10) (80) (13.0) (8.154) (0.075) 1-4 R-4 PAG-1 P-3 — Q-1 — — — (10) (80) (4.0) 1-5 R-5 PAG-2 P-3 — Q-2 — — — (10) (80) (13.0) 1-6 R-6 PAG-1 P-3 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-7 R-7 PAG-1 P-4 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-8 R-8 PAG-3 P-5 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-9 R-9 PAG-1 P-6 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-10 R-10 PAG-4 P-7 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-11 R-11 PAG-1 P-8 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-12 R-12 PAG-1 P-9 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-13 R-13 PAG-1 P-10 — Q-2 — FP-1 — (3) (80) (13.0) (3) 1-14 R-14 PAG-5 P-11 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-15 R-15 PAG-1 P-12 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-16 R-16 PAG-1 P-13 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-17 R-17 PAG-1 P-14 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-18 R-18 PAG-6 P-15 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-19 R-19 PAG-1 P-16 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-20 R-20 PAG-7 P-17 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-21 R-21 PAG-1 P-18 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-22 R-22 PAG-1 P-19 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-23 R-23 PAG-8 P-20 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-24 R-24 PAG-1 P-21 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-25 R-25 PAG-1 P-22 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-26 R-26 PAG-2 P-23 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-27 R-27 PAG-1 P-24 — Q-2 — FP-1 — (10) (80) (13.0) (3)

TABLE-US-00002 TABLE 2 Photoacid Fluorinated Resist generator Polymer Polymer Quencher Crosslinker polymer Surfactant composition (pbw) 1 (pbw) 2 (pbw) (pbw) (pbw) (pbw) (pbw)
Example 1-28 R-28 PAG-1 P-25 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-29 R-29 PAG-2 P-26 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-30 R-30 PAG-1 P-27 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-31 R-31 PAG-1 P-28 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-32 R-32 PAG-3 P-29 — Q-2 — FP-1 — (10) (80) (13.0) (3) 1-33 R-33 PAG-1 P-7 P-9 Q-1 — — — (8) (40) (40) (4.0) 1-34 R-34 PAG-1 P-7 P-9 Q-1 — FP-1 — (8) (40) (40) (4.0) (3) 1-35 R-35 PAG-4 P-7 P-9 Q-2 — — — (8) (40) (40) (13.0) 1-36 R-36 PAG-1 P-7 P-9 Q-2 — FP-1 — (8) (40) (40) (13.0) (3) 1-37 R-37 PAG-5 P-7 P-9 Q-3 — FP-1 — (8) (40) (40) (6.0) (3) 1-38 R-38 PAG-1 P-7 P-9 Q-3 TMGU FP-1 — (8) (40) (40) (6.5) (2.0) (3) 1-39 R-39 PAG-6 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-40 R-40 PAG-1 P-7 P-9 Q-4 — FP-2 — (8) (40) (40) (11.0) (5) 1-41 R-41 PAG-1 P-7 P-9 Q-4 — FP-3 — (8) (40) (40) (11.0) (1.5) 1-42 R-42 PAG-7 P-7 P-9 Q-4 — FP-4 — (8) (40) (40) (11.0) (5) 1-43

R-43 PAG-1 P-7 P-9 Q-4 — FP-5 — (8) (40) (40) (11.0) (5) 1-44 R-44 PAG-1 P-7 P-9 Q-4 — FP-1 — (20) (40) (40) (11.0) (3) 1-45 R-45 PAG-2 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-46 R-46 PAG-3 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-47 R-47 PAG-4 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-48 R-48 PAG-5 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-49 R-49 PAG-6 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-50 R-50 PAG-7 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-51 R-51 PAG-7 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-52 R-52 PAG-8 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-53 R-53 PAG-1 (8) P-7 P-9 Q-4 — FP-1 — PAG-A (2) (40) (40) (11.0) (3) 1-54 R-54 PAG-2 (8) P-30 P-9 Q-4 — FP-1 — PAG-A (2) (40) (40) (11.0) (3)

TABLE-US-00003 TABLE 3 Photoacid Fluorinated Resist generator Polymer Polymer Quencher Crosslinker polymer Surfactant composition (pbw) 1 (pbw) 2 (pbw) (pbw) (pbw) (pbw) (pbw) Comparative 1-1 CR-1 cPAG-1 P-1 — Q-1 TMGU — PF-636 Example (10) (80) (4.0) (8.154) (0.075) 1-2 CR-2 cPAG-1 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-3 CR-3 cPAG-2 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-4 CR-4 cPAG-3 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-5 CR-5 cPAG-4 P-7 P-9 Q-4 — FP-1 — (8) (40) (40) (11.0) (3) 1-6 CR-6 cPAG-1 P-3 — Q-2 — — — (10) (80) (13.0) 1-7 CR-7 cPAG-2 P-7 P-9 Q-4 — FP-3 — (8) (40) (40) (11.0) (1.5) 1-8 CR-8 cPAG-3 P-22 — Q-2 — FP-1 — (5) (80) (13.0) (3) 1-9 CR-9 cPAG-3 P-7 P-9 Q-3 — FP-1 — (8) (40) (40) (6.0) (3) 1-10 CR-10 cPAG-4 P-7 P-9 Q-3 TMGU FP-1 — (8) (40) (40) (6.5) (2.0) (3)

[0311] Polymers P-1 to P-30 used in resist compositions have the structure shown in Table 4.

TABLE-US-00004 TABLE 4 Incorpo- Incorpo- Incorpo- Incorpo- Incorpo- ration ration ration ration ration Unit ratio Unit ratio Unit ratio Unit ratio Unit ratio 1 (mol %) 2 (mol %) 3 (mol %) 4 (mol %) 5 (mol %) Mw Mw/Mn P-1 A-1 80.0 B-1 10.0 B-5 10.0 — — — — 4,500 1.65 P-2 A-1 80.0 B-2 8.0 B-4 12.0 — — — — 4,400 1.64 P-3 A-1 60.0 B-2 10.0 C-1 30.0 — — — — 3,700 1.62 P-4 A-1 70.0 B-2 7.0 C-2 23.0 — — — — 3,600 1.63 P-5 A-1 70.0 B-2 10.0 C-3 20.0 — — — — 3,900 1.65 P-6 A-1 30.0 B-2 10.0 C-4 60.0 — — — — 3,900 1.62 P-7 A-1 45.0 B-2 10.0 C-5 45.0 — — — — 4,100 1.63 P-8 A-1 55.0 B-4 10.0 C-1 35.0 — — — — 4,000 1.63 P-9 A-1 66.0 B-2 9.0 C-1 21.5 E-1 3.5 — — 13,000 1.62 P-10 A-1 60.0 B-2 4.0 C-1 24.0 E-1 12.0 — — 15,000 1.65 P-11 A-1 67.0 B-2 10.0 C-1 18.5 E-2 4.5 — — 14,000 1.63 P-12 A-1 67.0 B-2 9.3 C-1 20.0 E-3 3.7 — — 13,500 1.63 P-13 A-1 67.3 B-2 10.0 C-1 17.5 E-4 5.2 — — 13,200 1.64 P-14 A-1 64.1 B-2 9.5 C-1 22.0 E-5 4.4 — — 12,800 1.62 P-15 A-1 64.0 B-2 10.0 C-1 22.8 E-6 3.2 — — 13,500 1.63 P-16 A-1 62.0 B-3 10.0 C-1 24.3 E-1 3.7 — — 12,400 1.66 P-17 A-2 60.5 B-4 10.0 C-1 24.4 E-2 5.1 — — 12,300 1.65 P-18 A-1 70.0 C-1 30.0 — — — — — 4,200 1.69 P-19 A-1 80.0 B-2 5.0 C-1 15.0 — — — — 4,300 1.67 P-20 A-1 80.0 B-2 2.5 C-1 15.0 E-1 2.5 — — 12,100 1.69 P-21 A-2 50.0 C-1 30.0 D-1 20.0 — — — — 4,600 1.67 P-22 A-2 50.0 B-2 2.5 C-1 30.0 D-1 15.0 E-1 2.5 12,700 1.73 P-23 A-2 50.0 C-1 30.0 D-2 20.0 — — — — 5,400 1.72 P-24 A-2 50.0 C-1 30.0 D-3 20.0 — — — — 6,100 1.73 P-25 A-2 50.0 C-1 30.0 D-4 20.0 — — — — 7,000 1.76 P-26 A-1 67.5 B-2 2.5 C-1 30.0 — — — — 4,100 1.65 P-27 A-1 57.5 B-2 2.5 C-1 30.0 E-5 10 — — 11,000 1.65 P-28 A-1 65.0 C-1 25.0 E-7 10.0 — — — — 13,000 1.80 P-29 A-1 65.0 C-1 25.0 E-8 10.0 — — — — 12,500 1.80 P-30 A-3 76.0 B-2 6.0 C-1 18.0 — — — — 4,500 1.68

[0312] The structure of each unit in Table 4 is shown below.

##STR00665## ##STR00666## ##STR00667## ##STR00668## ##STR00669## ##STR00670##

[0313] In Tables 1 to 3, other photoacid generator PAG-A, comparative photoacid generators cPAG-1 to cPAG-4, quenchers Q-1 to Q-4, and fluorinated polymers FP-1 to FP-5 have the following structure.

##STR00671## ##STR00672## ##STR00673## ##STR00674##

[3] EB Lithography Test

Examples 2-1 to 2-54 and Comparative Examples 2-1 to 2-10

[0314] Using a coater/developer system ACT-M (Tokyo Electron Ltd.), each of the negative resist compositions (R-1 to R-54 and CR-1 to CR-10) was spin coated onto a mask blank of 152 mm

squares having the outermost surface of a silicon oxide film, which had been vapor primed with hexamethyldisilazane (HMDS), and prebaked on a hotplate at 110° C. for 600 seconds to form a resist film of 80 nm thick. The thickness of the resist film was measured by an optical film thickness measurement system Nanospec (Nanometrics Inc.). Measurement was made at 81 points in the plane of the blank substrate excluding a peripheral band extending 10 mm inward from the blank periphery, and an average film thickness and a film thickness range were computed therefrom.

[0315] The resist film was exposed to EB using an EB writer system EBM-5000Plus (NuFlare Technology Inc., accelerating voltage 50 kV), then baked (PEB) at 120° C. for 600 seconds, and developed in a 2.38 wt % TMAH aqueous solution, thereby yielding a negative pattern.

[0316] The resist pattern was evaluated as follows. The patterned mask blank was observed under a top-down scanning electron microscope (TDSEM). The optimum dose (Eop) was defined as the exposure dose ($\mu\text{C}/\text{cm}^2$) which provided a 1:1 resolution at the top and bottom of a 200-nm 1:1 line-and-space (LS) pattern. The resolution (or maximum resolution) was defined as the minimum line width of a LS pattern that could be resolved at the optimum dose.

[0317] The 200-nm LS pattern printed by exposure at the optimum dose (Eop) was observed under SEM. For each of the edges of 32 lines of the LS pattern, edge detection was carried out at 80 points, from which a 3-fold value (3 σ) of the standard deviation (σ) or variation was determined and reported as LER (nm). It was judged by visual observation whether or not the pattern profile was rectangular. For the evaluation of pattern fidelity, when a square dot pattern of size 120 nm and density 36% was placed, an area loss (%) at one corner of the square dot was computed. A smaller value indicates that the dot profile is more rectangular. The results are shown in Tables 5 to 7.

TABLE-US-00005 TABLE 5 Maximum Area Resist Eop resolution LER Pattern loss composition ($\mu\text{C}/\text{cm}^2$) (nm) (nm) profile (%) Example 2-1 R-1 100 38 4.9 rectangular 10 2-2 R-2 95 39 4.8 rectangular 10 2-3 R-3 100 40 4.8 rectangular 9 2-4 R-4 100 33 4.4 rectangular 10 2-5 R-5 105 35 4.3 rectangular 9 2-6 R-6 100 35 4.3 rectangular 9 2-7 R-7 105 34 4.2 rectangular 9 2-8 R-8 100 35 4.2 rectangular 9 2-9 R-9 100 34 4.3 rectangular 9 2-10 R-10 100 34 4.2 rectangular 9 2-11 R-11 95 35 4.4 rectangular 9 2-12 R-12 95 36 4.3 rectangular 9 2-13 R-13 100 34 4.3 rectangular 9 2-14 R-14 100 35 4.2 rectangular 9 2-15 R-15 105 34 4.4 rectangular 9 2-16 R-16 100 34 4.2 rectangular 9 2-17 R-17 100 34 4.2 rectangular 9 2-18 R-18 100 35 4.2 rectangular 9 2-19 R-19 95 34 4.1 rectangular 9 2-20 R-20 100 33 4.3 rectangular 9 2-21 R-21 95 34 4.2 rectangular 9 2-22 R-22 95 35 4.3 rectangular 9 2-23 R-23 100 36 4.2 rectangular 9 2-24 R-24 95 34 4.3 rectangular 9 2-25 R-25 100 34 4.2 rectangular 9 2-26 R-26 100 35 4.2 rectangular 9 2-27 R-27 95 33 4.2 rectangular 9

TABLE-US-00006 TABLE 6 Maximum Area Resist Eop resolution LER Pattern loss composition ($\mu\text{C}/\text{cm}^2$) (nm) (nm) profile (%) Example 2-28 R-28 95 34 4.1 rectangular 9 2-29 R-29 105 33 4.2 rectangular 9 2-30 R-30 100 33 4.1 rectangular 9 2-31 R-31 95 33 4.2 rectangular 9 2-32 R-32 95 33 4.1 rectangular 9 2-33 R-33 100 34 4.2 rectangular 9 2-34 R-34 95 34 4.2 rectangular 9 2-35 R-35 100 33 4.3 rectangular 9 2-36 R-36 105 34 4.2 rectangular 9 2-37 R-37 100 33 4.2 rectangular 9 2-38 R-38 95 34 4.2 rectangular 9 2-39 R-39 100 33 4.3 rectangular 9 2-40 R-40 105 32 4.1 rectangular 9 2-41 R-41 105 34 4.1 rectangular 9 2-42 R-42 105 33 4.2 rectangular 9 2-43 R-43 100 34 4.2 rectangular 9 2-44 R-44 105 33 4.2 rectangular 9 2-45 R-45 100 33 4.1 rectangular 9 2-46 R-46 105 33 4.2 rectangular 9 2-47 R-47 105 33 4.2 rectangular 9 2-48 R-48 105 34 4.3 rectangular 9 2-49 R-49 100 33 4.1 rectangular 9 2-50 R-50 95 33 4.2 rectangular 9 2-51 R-51 100 33 4.3 rectangular 9 2-52 R-52 105 32 4.2 rectangular 9 2-53 R-53 105 33 4.2 rectangular 9 2-54 R-54 105 33 4.1 rectangular 9

TABLE-US-00007 TABLE 7 Maximum Area Resist Eop resolution LER Pattern loss composition ($\mu\text{C}/\text{cm}^2$) (nm) (nm) profile (%) Comparative 2-1 CR-1 90 60 6.1 undercut 14 Example 2-2 CR-2 110 50 5.5 footing 15 2-3 CR-3 110 51 5.3 footing 16 2-4 CR-4 105 47 5.3 footing 15 2-5 CR-5 110 48 5.4 footing 15 2-6 CR-6 110 47 5.3 footing 16 2-7 CR-7 110 48 5.2 footing 14 2-8 CR-8 105 48 5.1 footing 15 2-9 CR-9 110 50 5.4 footing 16 2-10 CR-10 110 47 5.3 footing 15

[0318] All the chemically amplified negative resist compositions (R-1 to R-54) within the scope of the invention show satisfactory resolution, reduced LER, pattern rectangularity, and pattern fidelity. The comparative resist compositions (CR-1 to CR-10) lack resolution, LER and pattern rectangularity because the acid generator is of insufficient design.

[0319] The resist pattern forming process using the negative resist composition is useful in photolithography for the fabrication of semiconductor devices and the processing of photomask blanks of transmission or reflection type.

[0320] Japanese Patent Application No. 2024-031901 is incorporated herein by reference. Although some preferred embodiments have been described, many modifications and variations may be made thereto in light of the above teachings. It is therefore to be understood that the invention may be practiced otherwise than as specifically described without departing from the scope of the appended claims.

Claims

1. A chemically amplified negative resist composition comprising (A) a photoacid generator in the form of an onium salt having the formula (A) and (B) a base polymer containing a polymer comprising repeat units having the formula (B1), ##STR00675## wherein R^{sup.1} to R^{sup.12} are each independently hydrogen, halogen, or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, one of R^{sup.13} and R^{sup.14} is a group having a partial structure having the formula (a) and the other is hydrogen, halogen or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, at least two of R^{sup.1} to R^{sup.14} may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and Z^{sup.+} is an onium cation, ##STR00676## wherein m₁ is 0 or 1, m₂ is an integer of 0 to 4 when m₁=0 and an integer of 0 to 6 when m₁=1, m₃ is an integer of 0 to 3 when m₁=0 and an integer of 0 to 5 when m₁=1, m₂+m₃ is from 0 to 4 when m₁=0 and from 0 to 6 when m₁=1, R^{sup.F} is fluorine, a C_{sub.1}-C_{sub.6} fluorinated alkyl group, C_{sub.1}-C_{sub.6} fluorinated alkoxy group, or C_{sub.1}-C_{sub.6} fluorinated alkylthio group, R^{sup.15} is a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, a plurality of R^{sup.15} may bond together to form a ring with the carbon atoms to which they are attached when m₃ is 2 or more, L^{sup.A} and L^{sup.B} are each independently a single bond, ether bond, ester bond, amide bond, sulfonate ester bond, sulfonamide bond, carbonate bond or carbamate bond, and X^{sup.L} is a single bond or a C_{sub.1}-C_{sub.40} hydrocarbylene group which may contain a heteroatom, ##STR00677## wherein a₁ is 0 or 1, a₂ is 0, 1 or 2, a₃ is an integer meeting 0≤a₃≤5+2(a₂)-a₄, a₄ is 1, 2 or 3, R^{sup.A} is hydrogen, fluorine, methyl or trifluoromethyl, R^{sup.21} is halogen, an optionally halogenated C_{sub.1}-C_{sub.6} saturated hydrocarbyl group, optionally halogenated C_{sub.1}-C_{sub.6} saturated hydrocarbyloxy group, or optionally halogenated C_{sub.2}-C_{sub.8} saturated hydrocarbylcarbonyloxy group, and A^{sup.1} is a single bond or C_{sub.1}-C_{sub.10} saturated hydrocarbylene group in which some —CH_{sub.2}— may be replaced by —O—.

2. The negative resist composition of claim 1 wherein the onium salt has the formula (A^{sup.2}): ##STR00678## wherein R^{sup.5}, R^{sup.10} to R^{sup.13}, R^{sup.15}, L^{sup.A}, X^{sup.L}, R^{sup.F}, m₁ to m₃, and Z^{sup.+} are as defined above, m₄ and m₅ are each independently an integer of 0 to 4, R^{sup.16} and R^{sup.17} are each independently hydrogen, halogen, or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom; when m₄ is 2 or more, a plurality of R^{sup.16} may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and when m₅ is 2 or more, a plurality of R^{sup.17} may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon.

3. The negative resist composition of claim 1 wherein Z^{sup.+} is a sulfonium cation having the formula (cation-1) or iodonium cation having the formula (cation-2): ##STR00679## wherein

R.sup.ct1 to R.sup.ct5 are each independently halogen or a C.sub.1-C.sub.30 hydrocarbyl group which may contain a heteroatom, R.sup.ct1 and R.sup.ct2 may bond together to form a ring with the sulfur atom to which they are attached.

4. The negative resist composition of claim 1 wherein the polymer further comprises repeat units of at least one type selected from repeat units having the formula (B2) and repeat units having the formula (B3): ##STR00680## wherein b1 is 0 or 1, b2 is 0, 1 or 2, b3 is an integer meeting $0 \leq b3 \leq 5 + 2(b2) - b4$, b4 is 1, 2 or 3, b5 is 0, 1 or 2, b6 is 1 or 2, R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, R.sup.31 is halogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, R.sup.32 and R.sup.33 are each independently hydrogen, a C.sub.1-C.sub.15 saturated hydrocarbyl group which may be substituted with a hydroxy moiety or C.sub.1-C.sub.6 saturated hydrocarbyloxy moiety, or an optionally substituted C.sub.6-C.sub.15 aryl group, excluding that R.sup.32 and R.sup.33 are hydrogen at the same time, R.sup.32 and R.sup.33 may bond together to form a ring with the carbon atom to which they are attached, some —CH.sub.2— in the ring may be replaced by —O— or —S—, R.sup.34 is halogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, R.sup.35 and R.sup.36 are each independently hydrogen, a C.sub.1-C.sub.15 saturated hydrocarbyl group which may be substituted with a hydroxy moiety or C.sub.1-C.sub.6 saturated hydrocarbyloxy moiety, or an optionally substituted C.sub.6-C.sub.15 aryl group, excluding that R.sup.35 and R.sup.36 are hydrogen at the same time, R.sup.35 and R.sup.36 may bond together to form a ring with the carbon atom to which they are attached, some —CH.sub.2— in the ring may be replaced by —O— or —S—, A.sup.2 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group in which some —CH.sub.2— may be replaced by —O—, W.sup.1 and W.sup.2 are each independently hydrogen, a C.sub.1-C.sub.10 aliphatic hydrocarbyl group, C.sub.2-C.sub.10 aliphatic hydrocarbylcarbonyl group, or optionally substituted C.sub.6-C.sub.15 aryl group.

5. The negative resist composition of claim 1 wherein the polymer further comprises repeat units of at least one type selected from repeat units having the formula (B4), repeat units having the formula (B5), and repeat units having the formula (B6): ##STR00681## wherein c and d are each independently an integer of 0 to 4, e1 is 0 or 1, e2 is 0, 1 or 2, e3 is an integer of 0 to 5, R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, R.sup.41 and R.sup.42 are each independently hydroxy, halogen, an optionally halogenated C.sub.1-C.sub.8 saturated hydrocarbyl group, optionally halogenated C.sub.1-C.sub.8 saturated hydrocarbyloxy group, or optionally halogenated C.sub.2-C.sub.8 saturated hydrocarbylcarbonyloxy group, R.sup.43 is a C.sub.1-C.sub.20 saturated hydrocarbyl group, C.sub.1-C.sub.20 saturated hydrocarbyloxy group, C.sub.2-C.sub.20 saturated hydrocarbylcarbonyloxy group, C.sub.2-C.sub.20 saturated hydrocarbyloxyhydrocarbyl group, C.sub.2-C.sub.20 saturated hydrocarbylthiohydrocarbyl group, halogen, nitro, cyano, C.sub.1-C.sub.20 saturated hydrocarbylsulfinyl group, or C.sub.1-C.sub.20 saturated hydrocarbylsulfonyl group, and A.sup.4 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group in which some —CH.sub.2— may be replaced by —O—.

6. The negative resist composition of claim 4 wherein the polymer further comprises repeat units of at least one type selected from repeat units having the formulae (B7) to (B14): ##STR00682## ##STR00683## wherein R.sup.B is each independently hydrogen or methyl, Y.sup.1 is each independently a single bond, a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, naphthylene group or C.sub.7-C.sub.18 group obtained by combining the foregoing, *-O—Y.sup.11—, *-C(=O)—O—Y.sup.11—, or *-C(=O)—NH—Y.sup.11—, Y.sup.11 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, naphthylene group or C.sub.7-C.sub.18 group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, Y.sup.2 is each independently a single bond or *-Y.sup.21—C(=O)—O—, Y.sup.21 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom, Y.sup.3 is each independently a single bond, methylene, ethylene, phenylene, fluorinated phenylene, trifluoromethyl-substituted phenylene, *-O—Y.sup.31—, *-C(=O)—O—Y.sup.31—, or *-

C(=O)—NH—Y.sup.31—, Y.sup.31 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, fluorinated phenylene group, trifluoromethyl-substituted phenylene group, or C.sub.7-C.sub.20 group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, * designates a point of attachment to the carbon atom in the backbone, designates a point of attachment to the oxygen atom in the formula, Y.sup.4 is each independently a single bond or C.sub.1-C.sub.30 hydrocarbylene group which may contain a heteroatom, f1 and f2 are each independently 0 or 1, f1 and f2 are 0 when Y.sup.4 is a single bond, R.sup.51 to R.sup.68 are each independently halogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, R.sup.51 and R.sup.52 may bond together to form a ring with the sulfur atom to which they are attached, R.sup.53 and R.sup.54, R.sup.56 and R.sup.57, or R.sup.59 and R.sup.60 may bond together to form a ring with the sulfur atom to which they are attached, R.sup.HF is hydrogen or trifluoromethyl, and Xa.sup.— is a non-nucleophilic counter ion.

7. The negative resist composition of claim 6 wherein the polymer comprises repeat units having the formula (B1-1), repeat units having the formula (B2-1), (B2-2) or (B3-1), and repeat units having the formula (B8): ##STR00684## ##STR00685## wherein a4, b4, b6, R.sup.A, R.sup.B, Y.sup.2, R.sup.32, R.sup.33, R.sup.35, R.sup.36, R.sup.53, R.sup.54, R.sup.55, and R.sup.HF are as defined above.

8. The negative resist composition of claim 6 wherein the base polymer (B) further contains a polymer comprising repeat units having formula (B1) and repeat units having formula (B2) or (B3), but not repeat units having formulae (B7) to (B14).

9. The negative resist composition of claim 1 wherein repeat units having an aromatic ring structure account for at least 60 mol % of the overall repeat units of the polymer in the base polymer.

10. The negative resist composition of claim 1, further comprising (C) a crosslinker.

11. The negative resist composition of claim 4 which is free of a crosslinker.

12. The negative resist composition of claim 1, further comprising (D) a fluorinated polymer comprising repeat units of at least one type selected from repeat units having the formula (D1), repeat units having the formula (D2), repeat units having the formula (D3) and repeat units having the formula (D4) and optionally repeat units of at least one type selected from repeat units having the formula (D5) and repeat units having the formula (D6): ##STR00686## wherein x is an integer of 1 to 3, y is an integer satisfying $0 \leq y \leq 5 + 2z - x$, z is 0 or 1, g is an integer of 1 to 3, R.sup.C is each independently hydrogen, fluorine, methyl or trifluoromethyl, R.sup.D is each independently hydrogen or methyl, R.sup.101, R.sup.102, R.sup.104 and R.sup.105 are each independently hydrogen or a C.sub.1-C.sub.10 saturated hydrocarbyl group, R.sup.103, R.sup.106, R.sup.107 and R.sup.108 are each independently hydrogen, a C.sub.1-C.sub.15 hydrocarbyl group, C.sub.1-C.sub.15 fluorinated hydrocarbyl group, or acid labile group, and when R.sup.103, R.sup.106, R.sup.107 and R.sup.108 each are a hydrocarbyl or fluorinated hydrocarbyl group, an ether bond or carbonyl moiety may intervene in a carbon-carbon bond, R.sup.109 is hydrogen or a C.sub.1-C.sub.5 straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond, R.sup.110 is a C.sub.1-C.sub.5 straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond, R.sup.111 is a C.sub.1-C.sub.20 saturated hydrocarbyl group in which at least one hydrogen is substituted by fluorine, and in which some-CH.sub.2— may be replaced by an ester bond or ether bond, Z.sup.1 is a C.sub.1-C.sub.20 (g+1)-valent hydrocarbon group or C.sub.1-C.sub.20 (g+1)-valent fluorinated hydrocarbon group, Z.sup.2 is a single bond, *—C(=O)—O— or *—C(=O)—NH—, * designates a point of attachment to the carbon atom in the backbone, Z.sup.3 is a single bond, —O—, *—C(=O)=O—Z.sup.31—Z.sup.32— or *—C(=O)—NH—Z.sup.31—Z.sup.32—, Z.sup.31 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group, Z.sup.32 is a single bond, ester bond, ether bond, or sulfonamide bond, and * designates a point of attachment to the carbon atom in the backbone.

13. The negative resist composition of claim 1, further comprising (E) a quencher.

- 14.** The negative resist composition of claim 13 wherein the acid generator (A) and the quencher (E) are present in a weight ratio of less than 6/1.
- 15.** The negative resist composition of claim 1, further comprising (E) an organic solvent.
- 16.** A resist pattern forming process comprising the steps of: applying the chemically amplified negative resist composition of claim 1 onto a substrate to form a resist film thereon, exposing the resist film patternwise to high-energy radiation, and developing the exposed resist film in an alkaline developer.
- 17.** The process of claim 16 wherein the high-energy radiation is EUV of wavelength 3 to 15 nm or EB.
- 18.** The process of claim 16 wherein the substrate has the outermost surface of a material containing at least one element selected from chromium, silicon, tantalum, molybdenum, cobalt, nickel, tungsten, and tin.
- 19.** The process of claim 16 wherein the substrate is a mask blank of transmission or reflection type.
- 20.** A mask blank of transmission or reflection type which is coated with the chemically amplified negative resist composition of claim 1.
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